

Physics-Based State Reconstruction and Event-Source Diagnosability With Sparse PMU Measurements

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ABSTRACT Recovering unmeasured bus voltages and identifying a disturbance source are different inverse problems: sparse PMUs may determine a voltage function while leaving two sources indistinguishable. We separate them with a hybrid differential-algebraic model and specialize functional observability and descriptor-model distinguishability to the requested decisions. When nuisance cancels an event tangent, relative curvature can retain second-order evidence; this mechanism remains untested on the IEEE 39-bus system. Three frozen contracts test distinct parts of the formulation. A causal PowerDynamics estimator reaches 0.009634% mean hidden-voltage error, and measurement-only physical recentering removes 97.4%–98.3% of the operating-point error gap. A matched, known-onset, fixed-topology experiment ranks a closed bank of 16 load sources without source-labeled training. Source Top-1 falls from 100% for a 0.4% load change to 13.0% at 0.01%, although the latter remains detectable; source-wise response information is associated with the discretized 90%-detection threshold ($\rho = -0.794$, $n = 16$). In a separate ANDES bank, a supervised hierarchy localizes only 3 of 303 decisions after complete source holdout. The results separate state recoverability, event detectability, source resolution, and learned transfer. They do not validate multi-family nuisance-projected localization, unknown onset, open-bank events, changing PMU sets, ambient load variation, or field transfer.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

PHASOR measurement units (PMUs) reveal electromechanical transients through synchronized voltage and current phasors, yet transmission networks instrument only a fraction of their buses and branches [1], [2]. A disturbance may begin outside the measured region, alter hidden voltages, and propagate to the available devices. Missing frames and corrupted channels create a second ambiguity because they can resemble physical transients. Diagnosis must therefore separate three questions: what is the network state, which event occurred, and where did each applicable physical or measurement event originate?

Dynamic estimators combine network models with synchrophasors [2], [3]. Sparse-PMU classifiers identify events and faulted assets [4], [5]. Other learning methods reconstruct hidden trajectories before localization or estimate state and attack location from partial observations [6], [7]. These

methods solve different inverse problems. The unresolved case considered here has an under-observed state, concurrent physical and measurement-integrity changes, source assets absent from labeled training, and sensing geometries that may render two sources indistinguishable.

The distinction is operational, not semantic. A generation change can be visible throughout the network while its source remains ambiguous. A dropped PMU identifies an integrity location directly but removes information needed to localize a simultaneous physical event. Likewise, a state estimate can have small point error while the intervention that produced it is not uniquely identifiable. A single multiclass label conceals these differences and produces accuracy values whose denominators do not match the claimed task.

A physical candidate is represented by $h = (c, \ell, \tau, \gamma)$, containing the event family, source, onset, and magnitude. Measurement availability and corruption remain separate

variables. Each candidate generates a counterfactual DAE response, while candidate comparison fits admissible initial-state and integrity nuisance directions. Specializing established distinguishability and matched-subspace ideas [8], [9], the remaining weighted distance measures local source separation. A zero distance identifies a pair for which a uniformly correct source decision is impossible under the declared local model.

Three frozen contracts supply complementary evidence. The PowerDynamics state study reconstructs 31 hidden complex bus voltages from eight voltage/current PMUs and tests operating-point mismatch. A second PowerDynamics contract uses candidate-specific nonlinear load interventions to infer event presence, amplitude, and one of 16 source buses without a fitted source-label head. The ANDES study evaluates a physics-guided hierarchical ExtraTrees baseline on eight abnormal labels plus normal operation. The two simulators have different semantics, state coordinates, and data splits; their metrics are never pooled.

The contributions are:

- 1) a hybrid DAE inference model that maps eight event labels to composable physical-intervention and measurement-integrity variables;
- 2) a finite-horizon analysis that separates target-state estimability from event-source distinguishability and characterizes first- versus second-order evidence relative to admissible nuisance; and
- 3) frozen tests of hidden-voltage reconstruction, operating-point recentering, and Bayesian load-source inference, accompanied by a complete-source-holdout audit of the earlier supervised hierarchy.

The empirical boundary is sharp. The state estimator reaches 0.009634% mean hidden-voltage HVRE, while physical recentering closes 97.4%–98.3% of the operating-point error gap. In the load-intervention study, the physical likelihood resolves all sources at 0.4% magnitude but only 13.0% at 0.01%. The supervised event model detects and classifies known-source disturbances, yet complete source holdout leaves 3 correct decisions among 303. Physical candidate inference is therefore supported for one matched event family, not for the full eight-family joint problem.

II. PRIOR WORK AND RESEARCH GAP

A. STATE, ALGEBRAIC VARIABLE, AND UNKNOWN-INPUT ESTIMATION

Power-system dynamic state estimation combines machine, control, load, and network models with synchronized measurements; model uncertainty and measurement availability remain persistent constraints [2]. Robust filters address non-Gaussian measurement errors [10], [11], whereas joint input/state estimators recover exogenous inputs in linear systems [12], nonlinear DAEs [3], [13], and event-triggered or distributed power networks [14], [15]. These estimators do not define a multi-family physical-source bank with a simultaneous integrity state.

Learning methods reconstruct states from sparse measurements by encoding graph structure or power-flow relations. Examples combine graph learning with physics-based estimates [16], outlier-resistant estimation and outage identification [17], or topology-aware reconstruction under PMU failures and bad data [18]. Robust reconstruction does not, by itself, identify the physical source.

B. SPARSE-PMU EVENT AND SOURCE INFERENCE

PMU event identification uses subspaces, engineered features, transfer learning, and graph models [25]–[28]. Label-light field-stream detection and scarce-label transfer target event presence or family [29], [30]; physical-source recovery is distinct. Prior localization spans joint classification/localization [4], sparse-PMU fault, generation-loss, and outage methods [5], [31]–[33], group-sparse physical dictionaries [34], and topology-constrained graph losses [35].

Model-based outage inference provides the closest classical comparison. Phasor angles support line-outage detection [19]; uncertain pre-event state can be marginalized jointly with outage identity and sensor placement [20]; and sequential likelihoods handle stochastic operating variation [21], [36]. Matched-subspace detectors profile linear nuisance through a generalized likelihood ratio [9]. These are the likelihood and nuisance-elimination precedents used here. Our load contract instead uses a sparse voltage/current response dictionary and known onset; it is not a quickest-detection comparison.

Open-set recognition differs from source holdout. Siamese boundaries and generative evidence reject events outside learned classes [37]–[39]. Source generalization instead ranks a physically enumerable but unlabeled asset as a counterfactual source.

The pmuBAGE collection supports standardized analytics but does not certify source holdout [40]. Integrity is a separate variable: robust recovery distinguishes excursions from outliers [41], while utility examples include missing, locked, quantized, and timing-affected synchrophasors [42].

The unresolved issue is the evaluation contract. New replicas from trained assets measure interpolation, not source generalization; nuisance variation can also alter conclusions [43]. Reconstruction followed by localization is prior art [6], [7], but not under the combined held-source, diagnosability, integrity, and mismatch tests used here.

C. OBSERVABILITY, DIAGNOSABILITY, AND SENSOR VALUE

Functional observability asks whether a requested function of state can be inferred without full-state observability [44], [45]; nonlinear extensions require the observation fibers to determine the requested functional locally, not merely a favorable Jacobian at one point [46]. Unknown-input observability instead asks whether exogenous inputs can be reconstructed from output histories [22]; fault detectability asks whether a fault can be separated while disturbances are decoupled [47]. Sparse strong observability extends the question to corrupted

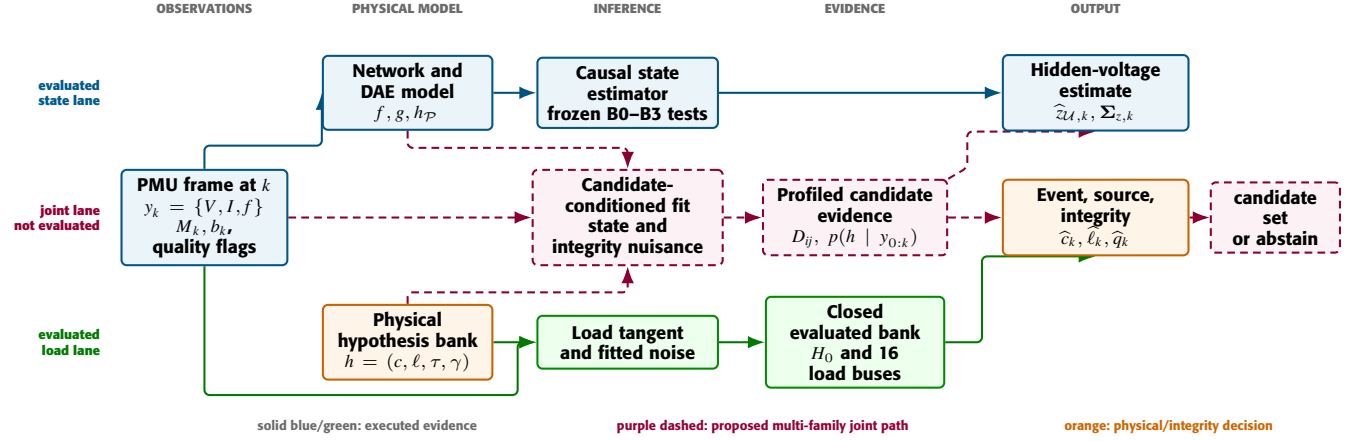


FIGURE 1. Inference path and validation boundary. The three lanes separate the evaluated causal state estimator, the proposed joint candidate-conditioned route, and the evaluated load-source special case. Solid blue and green boxes denote executed evidence; purple dashed boxes denote the multi-family nuisance fit, candidate comparison, and abstention path that still require a common experiment.

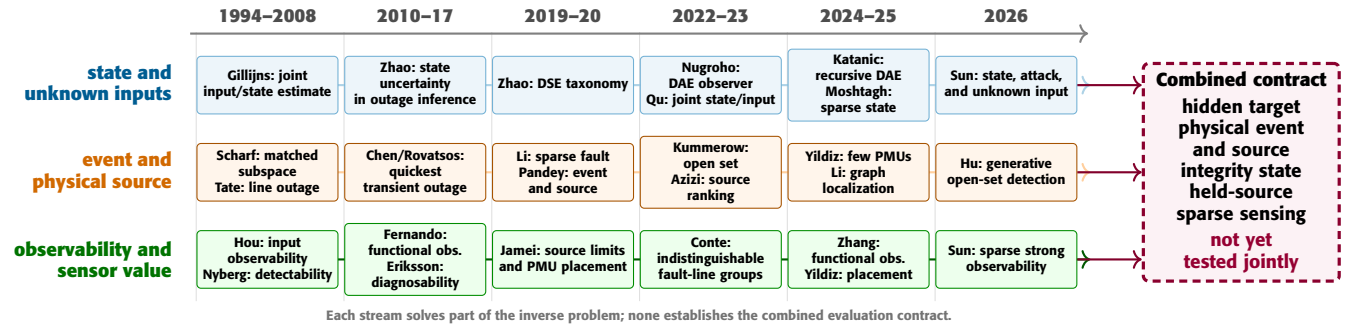


FIGURE 2. Three research streams converge on the unresolved contract. State/input estimators reconstruct latent variables [12]–[14]; event methods assign a family or physical source [4], [19]–[21]; observability and diagnosability determine whether the requested quantity is identifiable from the sensors [8], [22]–[24]. The dashed target requires all three capabilities under source-disjoint sparse sensing and a separate integrity state.

sensor sets [48], [49], and attack identification characterizes when adversarial input supports are distinguishable from system behavior [23]. Eriksson *et al.* quantify pairwise diagnosability for stochastic linear descriptor models [8]. Likelihood inference with singular first-order information also has established nonregular asymptotics [50]; therefore, a vanishing projected tangent is not by itself a universal impossibility result. PMU placement based on empirical observability Gramians treats sensor value as task dependent [51], [52]. For event sources, Jamei *et al.* derive placement-dependent fault-localization limits [24]; Conte *et al.* partition a radial grid into indistinguishable line clusters [53]; and Yildiz and Abur optimize sparse placement for event detection and identification [54]. These results distinguish hidden-state reconstruction, input reconstruction, and pairwise source separation. They also rule out a general optimal-placement claim from one prescribed eight-PMU set.

D. REMAINING TECHNICAL GAP

Table 1 separates the capabilities needed for the combined problem. A source-agnostic physical hypothesis must be evaluated while the unobserved trajectory is estimated and

measurement failures remain possible. The sensing geometry must expose indistinguishable sources before a classifier is trusted. Evaluation must then exclude the physical source asset from all source-labeled fitting. Under this protocol, learning is useful only when it corrects model discrepancy or reduces candidate-search cost without memorizing asset labels.

The available evidence covers only part of this intersection. PowerDynamics supports hidden-voltage estimation and physical recentering. ANDES supports known-source event recognition and documents failure under source holdout. Joint candidate-conditioned inference and the predicted relation between separation and uncertainty still require a common experimental contract.

III. JOINT STATE, EVENT, AND SOURCE PROBLEM

A. HYBRID DAE AND ELECTRICAL MEASUREMENT MODEL

Let $x(t) \in \mathbb{R}^{n_x}$ denote differential states, $z(t) \in \mathbb{R}^{n_z}$ algebraic states, $u(t)$ scheduled inputs, $q(t)$ network configuration, and $\theta_k = \theta_0 + G_\alpha \alpha_k$ physical parameters with a slowly varying operating coordinate α_k . A physical intervention $a_h(t)$ is

TABLE 1. Capability comparison for the closest methodological families. “Held source” means that the physical asset is absent from source-labeled training or tuning. Entries describe published tasks and assumptions, not a cross-paper performance ranking.

Work	State target	Event/source target	Physics mechanism	Sparse/partial sensing	Held source	Integrity separate
Katanic <i>et al.</i> [3]	Differential and algebraic DAE states	Unknown injectors, not event taxonomy	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Nugroho <i>et al.</i> [13]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance, not compositional source
Qu <i>et al.</i> [14]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo <i>et al.</i> [16]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue <i>et al.</i> [17]	Distribution-system state	Line outage	Grid-model-informed graph learning and outlier-resistant SE	Yes	Not stated	No
Moshtagh <i>et al.</i> [18]	Transmission-system state	Topology variation, not source output	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad-data tests
Pandey <i>et al.</i> [4]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [5]	None	Fault detection and location	OLS/CNN with strategic PMU placement	Yes	Not stated	No
Li <i>et al.</i> [35]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow <i>et al.</i> [38]	None	Known/unknown event type and location	Siamese distance with learned rejection	Limited/full sets	PMU	Open-set rejection
Hu and Cheng [39]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No physical-source posterior
Rimkus <i>et al.</i> [6]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	No source labels for imputer	Corruption via imputation
Zhai <i>et al.</i> [7]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmark contracts	Attack target only
This paper	Hidden voltage field; state uncertainty	Eight event labels; physical and integrity sources	DAE hypotheses; one-family physical likelihood	Eight prescribed PMU buses	Yes; learned baseline fails	Yes

indexed by hypothesis

$$h = (c, \ell, \tau, \gamma, d), \quad h \in \mathcal{H}, \quad (1)$$

where c is event family, ℓ is a bus or branch source, τ is onset, γ is severity, and d describes the allowed temporal profile. The plant is

$$\dot{x}(t) = f(x, z, u, q, a_h; \theta_k) + w_x(t), \quad (2)$$

$$\mathbf{0} = g(x, z, u, q, a_h; \theta_k), \quad (3)$$

$$y_k = M_k[h\mathcal{P}_k(x_k, z_k, q_k; \theta_k) + b_k] + v_k. \quad (4)$$

Here M_k selects the channels that are available at time k , b_k is a measurement-corruption process, and v_k is measurement noise. A dropout changes M_k ; bad data change b_k ; neither is a physical input to (2)–(3). This separation permits a physical event and a measurement failure to coexist.

The latent variables have deliberately different time scales and meanings. The pair (x, z) is the fast dynamic–algebraic state; α is slow operating-point or parameter drift; a_h is an abrupt, sparse physical intervention; q is a discrete network mode; and (M, b) describes data availability and corruption. A colored model-discrepancy state c_k may enter either the process or measurement residual, as tested in Section V, but is not interchangeable with an event.

Within a smooth mode, assume the DAE is locally index one, so g_z is nonsingular after the reference-angle constraint is imposed. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1} \left(g_x f + g_u \dot{u} + g_\theta \dot{\theta} \right). \quad (5)$$

Equation (5) is not applied across a fault insertion or topology switch, where algebraic variables can jump and the model must be segmented. Differential states remain continuous unless a simulator-specific reset map is declared.

The algebraic network core is

$$I = Y_{\text{bus}}(q, \theta)V, \quad S_i = P_i + jQ_i = V_i I_i^*. \quad (6)$$

Writing $Y_{\text{bus}} = G + jB$ and $V_i = |V_i|e^{j\vartheta_i}$ gives

$$P_i = |V_i| \sum_j |V_j| [G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}], \quad (7)$$

$$Q_i = |V_i| \sum_j |V_j| [G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}], \quad (8)$$

where $\vartheta_{ij} = \vartheta_i - \vartheta_j$. A line outage changes the branch primitives and therefore Y_{bus} ; a bus fault adds a shunt-admittance intervention; generation and load changes modify the appropriate active/reactive injection equations. For branch $e = (i, j)$, let t_e be the complex off-nominal tap, y_e the series

admittance, and $y_e^{f,sh}, y_e^{t,sh}$ the terminal shunts. Its terminal currents obey

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \quad (9)$$

The PMU operator selects the actual instrumented terminal and orientation. Replacing terminal current with complete bus-injection current, or changing the tap and charging convention, changes the likelihood.

At the acquisition boundary, native measurements can include V_{abc} , I_{abc} , frequency, ROCOF, quality flags, and availability. Positive-sequence phasors define the balanced network likelihood used here. Negative- and zero-sequence components, when available, remain unbalance or integrity evidence rather than being mixed into the positive-sequence state. Ideal continuous-frequency channels satisfy

$$f_i = f_0 + \frac{1}{2\pi} \frac{d\angle V_i}{dt}, \quad \text{ROCOF}_i = \frac{1}{2\pi} \frac{d^2\angle V_i}{dt^2}, \quad (10)$$

but an implemented likelihood must use the causal estimator and filtering semantics of the corresponding PMU backend.

B. SPARSE PMUS AND REQUESTED STATE FUNCTION

Both evidence campaigns use PMU buses

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}, \quad |\mathcal{P}| = 8. \quad (11)$$

The unobserved set is $\mathcal{U} = \{1, \dots, 39\} \setminus \mathcal{P}$, with $|\mathcal{U}| = 31$. The state-estimation campaign stacks rectangular voltage and controlled terminal-current components,

$$y_{\mathcal{P},k} = \text{col}_{p \in \mathcal{P}} [\Re V_{p,k}, \Im V_{p,k}, \Re I_{p\ell,k}, \Im I_{p\ell,k}], \quad (12)$$

and requests the hidden complex-voltage function

$$z_{\mathcal{U},k} = \text{col}_{i \in \mathcal{U}} [\Re V_{i,k}, \Im V_{i,k}] \in \mathbb{R}^{62}. \quad (13)$$

The event campaign also uses voltage angle, frequency, availability, and, in its longer-window implementation, current and ROCOF. Measurement semantics are kept campaign-specific rather than declaring channels equivalent across simulators.

C. EIGHT EVENT LABELS AS COMPOSABLE STATES

Table 2 defines the eight abnormal labels in the historical event bank. The scientific variables are not the label numbers. Classes 1–4 are single physical interventions; classes 5 and 7 are measurement-integrity states; classes 6 and 8 are compositions. Class 8 is a prescribed mixed event in the available simulations, not evidence of open-set recognition.

Table 3 states how the four physical families enter the candidate DAE. It defines the operators without implying that the joint posterior has already solved every family.

The internal decision at time k is

$$(\hat{z}_{\mathcal{U},k}, \Sigma_{z,k}, \hat{c}_k, \hat{\ell}_k^{\text{phys}}, \hat{m}_k, \hat{\ell}_k^{\text{meas}}, \mathcal{C}_k), \quad (14)$$

where m_k denotes the integrity state and $\mathcal{C}_k \subseteq \mathcal{H}$ is a credible candidate set. A location is empty when it does not apply. The candidate set is required because limited observability can make a forced Top-1 decision scientifically unjustified.

D. CORRECTNESS AND EVALUATION UNITS

State accuracy uses the mean hidden-voltage relative complex error (HVRE)

$$\text{HVRE}^{(r)} = \frac{1}{T_r |\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\hat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \quad (15)$$

This trajectory-and-bus aggregate is reported as a percentage. It is TVE-style but is not the standard per-frame synchrophasor TVE conformity statistic. Event detection, event-family macro-F1, exact physical Top-1/Top-3, and integrity-source Top-1 use their own declared populations. The joint correctness indicator is one only when every applicable component of (14) is correct; a correct event family cannot conceal a wrong source. Confidence intervals use an independently generated trajectory, event parent, or rebuilt plant as the statistical unit, never individual 30-Hz samples.

Causal inference permits only $y_{0:k}$ at time k . A fixed-lag smoother, complete-window statistic, or centered temporal feature is retrospective and is labeled accordingly. Source generalization requires that physical source ℓ be absent from all source-labeled training and tuning, not merely that the test trajectory use a new random seed.

IV. RECOVERABILITY AND DIAGNOSABILITY UNDER SPARSE PMUS

A. FUNCTIONAL OBSERVABILITY OF THE REQUESTED VOLTAGE FIELD

Linearizing the index-reduced DAE around a verified equilibrium and sampling at $\Delta t = 1/30$ s gives

$$\delta x_{k+1} = A_d \delta x_k + w_k, \quad (16)$$

$$\delta y_k = C \delta x_k + v_k, \quad (17)$$

$$\delta z_{\mathcal{U},k} = L_{\mathcal{U}} \delta x_k. \quad (18)$$

The reduction assumes a regular local algebraic Jacobian after the reference mode is fixed. Here $\delta x_k \in \mathbb{R}^{114}$ contains reduced differential coordinates, C maps them to 32 rectangular voltage/current PMU channels, and $L_{\mathcal{U}}$ maps them to 62 rectangular hidden-voltage components. These matrices are local and hypothesis dependent after a topology change.

For a horizon of N transitions, define

$$\mathcal{O}_N = \text{col}(C, CA_d, \dots, CA_d^N). \quad (19)$$

Full-state observability requires $\text{rank}(\mathcal{O}_N) = 114$. The requested voltage at time $r \leq N$ instead has map $L_r = L_{\mathcal{U}} A_d^r$. Recovering the complete local target sequence uses $\mathcal{L}_N = \text{col}(L_0, \dots, L_N)$.

Proposition 1 (Finite-horizon functional reconstruction): For the noise-free local system, the target at time r is uniquely determined by the output sequence through N if and only if

$$\ker \mathcal{O}_N \subseteq \ker L_r. \quad (20)$$

The complete target sequence is uniquely determined if and only if $\ker \mathcal{O}_N \subseteq \ker \mathcal{L}_N$.

Proof: Two initial states produce the same output history exactly when their difference belongs to $\ker \mathcal{O}_N$. Their

TABLE 2. Event ontology and historical ANDES inventory. Each source configuration has five replicas. “Physical source” and “integrity source” are separate outputs.

ID	Operational meaning	Physical intervention c	Physical source ℓ	Integrity state/source	Records
0	Normal operation	none	none	nominal	5
1	Bus fault	fault admittance	bus	nominal	195
2	Line outage	topology removal	branch	nominal	170
3	Generation change	active/reactive injection	generator bus	nominal	50
4	Load change	active/reactive withdrawal	load bus	nominal	90
5	Missing data	none	none	dropout / PMU	40
6	Generation change with missing data	generation change	generator bus	dropout / PMU	50
7	Bad data	none	none	corruption / PMU channel	40
8	Prescribed mixed disturbance	generation plus load changes	declared generator bus	measurement disturbance / PMU	50
Total					690

TABLE 3. Physical intervention operators in the candidate bank. Algebraic jumps are resolved by the post-event network constraints; differential-state resets are absent unless explicitly stated.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive, $Y_{bus}^+ = Y_{bus}^- + y_f e_\ell e_\ell^T$, for the declared clearing interval	admissible buses	complex fault admittance or resistance/reactance	x continuous; z may jump
Line outage	Set branch status $q_{ij} : 1 \mapsto 0$ and remove its tap, series, charging, and shunt primitives from Y_{bus}	in-service branches	discrete status and switching time	x continuous; z may jump
Generation change	Modify scheduled active/reactive injection or a declared governor reference at generator bus ℓ	generator buses	$\Delta P_G, \Delta Q_G$, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model at load bus ℓ	load buses	$\Delta P_L, \Delta Q_L$, ZIP convention	continuous input-driven response

requested outputs at time r agree for every such pair exactly when the difference also belongs to $\ker L_r$. Stacking the maps gives the sequence statement. Equivalently, the rows of the requested map lie in the row space of $\mathcal{O}_N$. ■

This standard functional-observability result [44], [45] fixes the target and time index before any estimator is assessed. It also shows why a low full-state rank alone does not preclude recovery of the hidden-voltage function. In a nonlinear DAE, however, a favorable kernel inclusion at one equilibrium is only first-order evidence: local functional factorization requires a constant-rank neighborhood condition on the feasible observation fibers [46].

The Gauss–Markov specialization adds a variance statement to the rank condition. Stack the output perturbations as $\delta Y_N = \mathcal{O}_N \delta x_0 + v_N$, assume $v_N \sim \mathcal{N}(\mathbf{0}, R_N)$ with $R_N \succ 0$, and define $G_N = \mathcal{O}_N^T R_N^{-1} \mathcal{O}_N$.

Corollary 1 (Minimum-variance functional estimate): If (20) holds, the best linear unbiased estimate of the time- r target and its error covariance are

$$\widehat{\delta z}_{\mathcal{U},r} = L_r G_N^\dagger \mathcal{O}_N^T R_N^{-1} \delta Y_N, \quad (21)$$

$$\Sigma_{z,r}^* = L_r G_N^\dagger L_r^T. \quad (22)$$

Every other linear unbiased estimator has covariance $\Sigma_{z,r}^* + QR_N Q^T$ for some Q satisfying $Q\mathcal{O}_N = \mathbf{0}$.

Proof: Condition (20) implies $L_r = L_r G_N^\dagger G_N$, which proves unbiasedness of (21). Direct substitution gives (22). Any other unbiased linear map is the displayed map plus Q with $Q\mathcal{O}_N = \mathbf{0}$. The covariance cross terms vanish because

$\mathcal{O}_N^T Q^T = \mathbf{0}$, leaving the positive-semidefinite term $QR_N Q^T$. ■

For whitened measurements, $\text{tr}(\Sigma_{z,r}^*)$ quantifies noise amplification in the requested voltage field. It depends on singular values and target alignment, not rank alone. The numerical curve below is an implementation of this quantity with a tolerance-dependent pseudoinverse; it is a conditioning diagnostic rather than an exact certificate.

B. NUMERICAL AUDIT FOR THE PRESCRIBED EIGHT-PMU CASE

For the anchor-time target used by the structural audit, define

$$\Pi_N = \mathcal{O}_N^\dagger \mathcal{O}_N, \quad (23)$$

$$r_F(N) = \|L_{\mathcal{U}}(I - \Pi_N)\|_F, \quad (24)$$

$$r_{F,i}(N) = \|L_i(I - \Pi_N)\|_F. \quad (25)$$

A zero residual is an exact numerical condition only when the matrix scaling and rank decision are defensible. A nonzero value certifies that the computed observable subspace does not contain every requested anchor-time direction.

The implementation formed $\mathcal{O}_N^\dagger \mathcal{O}_N$ and retained eigenvalues above 10^{-9} of its maximum. Figure 3 turns the audit into a claim test. The pseudo-rank reaches 54 at $N = 30$ but falls to 39 at $N = 180$, although the exact rank of a vertically augmented observability matrix cannot decrease. Gramian squaring, unstable-mode scaling, and a relative threshold therefore make this pseudo-rank unsuitable as an algebraic certificate. Longer history still lowers the trace-style noise bound by more than five orders of magnitude, but $r_F(N)$

remains nonzero. The evidence supports improved numerical conditioning, not full-state or exact target observability. Empirical voltage recovery is evaluated separately and is not used to repair the failed certificate.

Horizons $N \in \{0, 3, 10, 30, 60, 120, 180\}$ correspond to 0–6 s. The nonlinear nominal trajectories contain 91 frames, or 3 s, so $N = 180$ is not a matched causal horizon for those records. Its bus ranking was nevertheless exported before nonlinear scoring and is retained only as a preregistered, cross-horizon numerical diagnostic. The empirical test correlates $r_{F,i}(180)$ with bus-level HVRE; because both derive from the threshold-sensitive projector, that association is exploratory until an SVD-based, noise-tied rank audit is frozen. It does not convert the six-second local audit into a three-second observability proof.

C. EVENT-SOURCE DIAGNOSABILITY FROM CANDIDATE MANIFOLDS

State recoverability and source diagnosability have different nuisance spaces. The construction below specializes established descriptor-model distinguishability [8], matched-subspace nuisance profiling [9], and fault/attack identifiability [23], [47] to candidate-specific DAE responses and explicit integrity supports. The contribution is the common PMU problem contract and its separation from the requested-state problem, not the underlying projection identity. Under candidate h , stack the available channels over H samples as

$$Y_H = s_{h,H} + \mathcal{O}_{h,H}\xi_h + \mathcal{E}_{S,H}\beta_h + \varepsilon_H, \quad \varepsilon_H \sim \mathcal{N}(\mathbf{0}, R_H), \quad (26)$$

where $s_{h,H}$ is the intervention response, ξ_h is the locally admissible initial-state deviation, and $\mathcal{E}_{S,H}\beta_h$ is corruption on a fixed support S . Allowing each candidate to refit nuisance variables produces the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{col}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \quad (27)$$

This candidate-specific form is required for topology-changing events because $\mathcal{O}_{i,H}$ and $\mathcal{O}_{j,H}$ need not agree.

Let $\Delta s_{ij} = s_{i,H} - s_{j,H}$ and

$$\mathcal{N}_{ij,S} = [\mathcal{O}_{i,H}, -\mathcal{O}_{j,H}, \mathcal{E}_{S,H}]. \quad (28)$$

The R_H^{-1} -orthogonal nuisance projector and the pairwise separation margin are

$$P_{ij,S}^\perp = I - \mathcal{N}_{ij,S}(\mathcal{N}_{ij,S}^\top R_H^{-1} \mathcal{N}_{ij,S})^\dagger \mathcal{N}_{ij,S}^\top R_H^{-1}, \quad (29)$$

$$D_{ij,S}^{(\mathcal{P})}(H) = \frac{1}{2} \|P_{ij,S}^\perp \Delta s_{ij}\|_{R_H^{-1}}^2. \quad (30)$$

Equation (30) is half the minimum squared weighted distance between the two affine manifolds. It becomes the Gaussian mean-separation term only after nuisance values are fixed at the closest points. If all candidates are conditioned on one common pre-event state, the corresponding model replaces the two independent state blocks in (28) by $\mathcal{O}_{i,H} - \mathcal{O}_{j,H}$. The analysis below uses independent candidate-specific state blocks; it does not use the common-state alternative.

Proposition 2 (Local composite-hypothesis indistinguishability): If $D_{ij,S}^{(\mathcal{P})}(H) = 0$, then $\mathcal{M}_i(S)$ and $\mathcal{M}_j(S)$ intersect. Hence there exist admissible nuisance values under candidates i and j that induce the same Gaussian measurement law. No decision rule can distinguish the pair uniformly over the declared nuisance set; with equal prior mass on those two parameter points, its average error is at least $1/2$.

Proof: The margin is zero exactly when $\Delta s_{ij} \in \text{col}(\mathcal{N}_{ij,S})$, which is equivalent to an intersection of the two affine mean sets. At the intersecting nuisance values, the means and covariance are identical. The observation law is therefore identical under both parameter points, so the two conditional error probabilities sum to one. ■

When the corruption support is unknown but at most q channels are affected, the conservative margin is

$$D_{ij}^{(q)}(H) = \min_{S: |S| \leq q} D_{ij,S}^{(\mathcal{P})}(H). \quad (31)$$

This union-of-subspaces formulation connects source localization to attack identifiability, sparse strong observability, and unknown-input reconstruction [22], [23], [48], [49]. It also states the condition required for distinguishability under sparse corruption.

Corollary 2 (Monotonicity under an added PMU): For the same pair, covariance model, nuisance specification, and integrity budget, adding a conditionally independent PMU block cannot reduce the profiled margin:

$$D_{ij,S}^{(\mathcal{P} \cup \{p\})}(H) \geq D_{ij,S}^{(\mathcal{P})}(H). \quad (32)$$

For every nuisance vector, the augmented squared residual is the old nonnegative residual plus the new PMU residual. Taking the minimum preserves the inequality. Equality with a sum of separately profiled margins does not hold in general because the same nuisance variables must explain both sensor blocks; additivity requires fixed simple hypotheses or additional separability conditions.

The affine margin tests only first-order separation. To state what a zero margin does and does not imply, work in whitened coordinates and let the local no-event explanations and one event path be

$$F_0(b) = Nb + \frac{1}{2}H_0[b, b] + O(\|b\|^3), \\ c_g(a) = ad_g + a^2q_g + O(a^3). \quad (33)$$

Define $\Delta_g(a) = \min_{b \text{ local}} \|c_g(a) - F_0(b)\|_2$. Here b parameterizes a regular observable nuisance chart, so redundant coordinates have been removed and N has full column rank. The construction applies pairwise after one candidate manifold is used as the local reference.

Proposition 3 (Relative order of distinguishability): Let $P_\perp = I - N(N^\top N)^{-1}N^\top$. If $P_\perp d_g \neq \mathbf{0}$, then

$$\Delta_g^2(a) = a^2 \|P_\perp d_g\|_2^2 + O(a^3). \quad (34)$$

If $d_g = Nv$, define the relative curvature

$$\kappa_g = P_\perp \left(q_g - \frac{1}{2}H_0[v, v] \right). \quad (35)$$

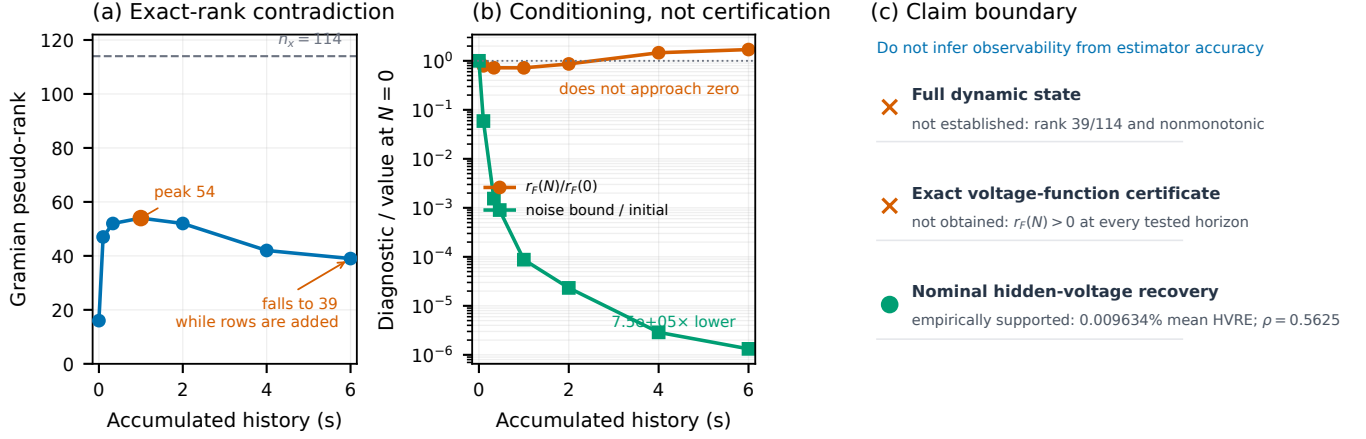


FIGURE 3. What the prescribed eight-PMU audit permits one to claim. (a) The relative-threshold Gramian pseudo-rank decreases after $N = 30$ even though samples are added, contradicting the monotonicity required of exact rank. (b) Values normalized to their $N = 0$ levels separate two effects: accumulated history sharply lowers the trace-style noise bound, whereas the functional residual does not approach zero. (c) Consequently, the audit does not establish the full dynamic state or an exact hidden-voltage function. The small nominal TEST error is empirical evidence under the frozen experiment, not an observability proof.

Then $\Delta_g^2(a) = a^4 \|\kappa_g\|_2^2 + O(a^5)$. Thus zero first-order information does not imply local nonlinear indistinguishability when $\kappa_g \neq 0$.

Proof: The stationarity equation has derivative $N^T N \succ 0$ in b , so the implicit-function theorem gives a smooth minimizer. With $v_* = (N^T N)^{-1} N^T d_g$, expanding $b_*(a) = av_* + a^2 w + O(a^3)$ leaves $aP_\perp d_g$ at first order. If $d_g = Nv_*$, profiling removes the tangential quadratic term and leaves $a^2 \kappa_g + O(a^3)$; squaring proves the claim. ■

For n independent Gaussian windows, the closest-null divergence is $n\Delta_g^2(a)/2$; Pinsker's inequality bounds level- α power by $\alpha + \min\{1, \sqrt{n}\Delta_g(a)/2\}$. Hence contact order r gives limiting amplitude scale $n^{-1/(2r)}$: $n^{-1/2}$ at first order and $n^{-1/4}$ at second order [50]. This result is local and mode preserving. Correlated frames enter through their joint covariance, not as independent replications.

A positive affine margin has a complementary guarantee. If each exact candidate response has a radius- r remainder bounded by $\kappa_h \|\eta_h\|_2^2/2$, the reverse triangle inequality gives $d_{ij}^{\text{nl}}(r) \geq [d_{ij}^{\text{lin}} - (\kappa_i + \kappa_j)r^2/2]_+$. Neither this bound nor Proposition 3 is certified for the multi-family bank; the load study measures only source-wise Taylor remainder.

The executed load-intervention case gives a narrower matched-signal specialization. Let D_g be the whitened unit-amplitude response for candidate source g and let Σ be the normal-data covariance before whitening. The amplitude Fisher information is

$$\mathcal{I}_g = D_g^T \Sigma^{-1} D_g, \quad |a|_{\min} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{\mathcal{I}_g}}, \quad (36)$$

where the second expression is the local amplitude required by a one-sided Gaussian score with false-alarm probability α and power $1 - \beta$. The formula predicts detectability, not source identity.

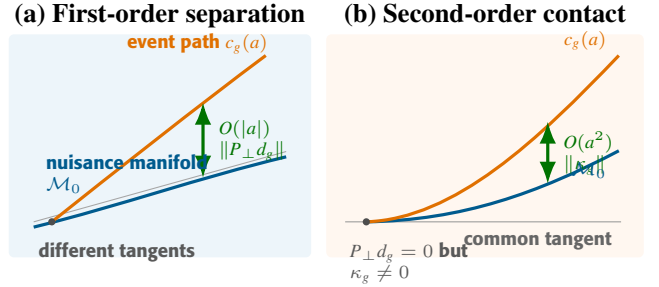


FIGURE 4. Local diagnosability after whitening. (a) A projected event tangent gives first-order separation. (b) Nuisance absorbs the tangent, but relative curvature yields squared separation proportional to a^4 . Event curvature alone is insufficient because the nuisance manifold can bend with it.

For source j as the closest single-signature alternative, the information left after its amplitude is fitted is

$$\mathcal{J}_{g \rightarrow j} = \mathcal{I}_g - \frac{(D_g^T \Sigma^{-1} D_j)^2}{D_j^T \Sigma^{-1} D_j} = \mathcal{I}_g (1 - \mu_{gj}^2), \quad (37)$$

with μ_{gj} the covariance-weighted signature coherence. Equation (37) is the profiled pairwise Fisher information for one signature against another. It need not predict multicandidate confusion once amplitude priors, unequal alternatives, and finite nonlinear response are present.

For a calibrated source posterior, define the smallest candidate set satisfying

$$\mathcal{C}_k(\alpha) = \arg \min_{\mathcal{C} \subseteq \mathcal{H}} \left\{ |\mathcal{C}| : \sum_{h \in \mathcal{C}} p(h | y_{0:k}) \geq 1 - \alpha \right\}. \quad (38)$$

A diagnosability-aware system abstains from exact Top-1 when $|\mathcal{C}_k| > 1$ or when its closest competing margin falls below a development-set threshold. The load-intervention study tests only the amplitude-information relation in (36) and reports an exploratory, mismatched aggregate comparison for

(37). It does not fit the initial-state nuisance $\mathcal{O}_h \xi_h$, corruption nuisance $\mathcal{E}_S \beta_h$, or the curvature constant κ_h . Candidate-set calibration, source-resolution delay, and PMU loss or addition still require a source-disjoint multi-family experiment.

V. PHYSICS-BASED STATE ESTIMATION AND JOINT INFERENCE

A. EXACT TARGET AND A GENERATIVE CANDIDATE MODEL

Let $\zeta_k = L(x_k, z_k)$ be the electrical quantity requested by the application. Under squared loss, the exact Bayesian target is

$$\hat{\zeta}_k^{\text{MMSE}} = \mathbb{E}[\zeta_k | y_{0:k}], \quad \hat{h}_k^{\text{MAP}} = \arg \max_{h \in \mathcal{H}} p(h | y_{0:k}). \quad (39)$$

This statement is an optimality definition, not an implemented-algorithm claim.

For candidate $h = (c, \ell, \tau, \gamma, d)$, assume $x_0 \sim \mathcal{N}(\bar{x}_0, P_0)$, Gaussian process and measurement noise, the hard constraint $g_h(x_t, z_t) = \mathbf{0}$, and a Laplace prior on active corruption coefficients. With

$$r_{h,t} = M_t[y_t - h p_t(x_t, z_t) - b_t], \quad (40)$$

the negative twice-log joint density conditional on h , up to constants and the declared corruption-support prior, is

$$\begin{aligned} \mathcal{J}_h^\circ = & \|x_0 - \bar{x}_0\|_{P_0^{-1}}^2 + \sum_{t=0}^k \|r_{h,t}\|_{R_t^{-1}}^2 \\ & + \sum_{t=0}^{k-1} \|x_{t+1} - F_h(x_t, z_t)\|_{Q^{-1}}^2 + 2\lambda_b \sum_{t=0}^k \|b_t\|_1, \end{aligned} \quad (41)$$

subject to the DAE constraints. Crucially, $\mathcal{J}_h^\circ$ does not contain the hypothesis prior. The physical family and source enter through F_h, g_h , and the network/measurement operators rather than as an asset-label head.

The evidence is the constrained integral

$$p(y_{0:k} | h) = \int_{g_h=0} \exp[-\mathcal{J}_h^\circ/2] dX dZ dB. \quad (42)$$

A Laplace approximation on locally identifiable directions gives

$$p(h | y_{0:k}) \propto p(h) \det^+(H_h)^{-1/2} \exp[-\mathcal{J}_h^\circ(\hat{X}_h, \hat{Z}_h, \hat{B}_h)/2], \quad (43)$$

so the prior appears exactly once. The pseudo-determinant requires the active constraint and corruption support to be fixed locally. If a Huber loss, a soft DAE penalty, or an omitted determinant is used without a corresponding normalized density, the result is a penalized evidence score, not a posterior probability.

The learned component is deliberately restricted. The earlier engineered feature system can propose a candidate set or learn a correction $\psi_\phi(r_{h,0:k})$ to physical log evidence. It must consume candidate residuals, not source identity alone, and must be tested on held-out physical sources. The required ablation is learned-only, physics-only, physics plus learned discrepancy, and the complete diagnosability-aware rule.

B. EVALUATED BAYESIAN LOAD-SOURCE MODEL

One restricted instance of the candidate model is executed rather than proposed. For each of 16 load buses, a centered finite difference of the nonlinear time-domain simulator forms a 30-frame, 32-channel response dictionary,

$$D_g = \frac{y_g(+\epsilon) - y_g(-\epsilon)}{2\epsilon}, \quad \epsilon = 0.005. \quad (44)$$

The dictionary is physical: it is regenerated by changing the load at candidate bus g and simulating the network. No source classifier or source label is fitted. Its frozen SHA-256 identifier is recorded in the repository manifest.

After subtracting the normal mean, the first-order event model is

$$r = aD_g + \epsilon, \quad a \sim \mathcal{N}(0, \sigma_a^2), \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma). \quad (45)$$

The covariance is estimated from equilibrium records with synthetic measurement noise, not stochastic load trajectories. Gaussian innovations use standard deviations 2×10^{-4} and 5×10^{-4} p.u. for voltage and current coordinates, AR(1) coefficient 0.35, fixed cross-channel coupling, and disjoint CAL/DEV/TEST seeds. Ambient load variation, instrument/timing error, and noise sweeps are absent, so weak-event thresholds apply only to this generator. W0 is an unscaled diagnostic, W1 estimates channel covariance, and W2 adds separable first-order temporal correlation. Under (45), the conditional amplitude posterior is Gaussian,

$$s_g^2 = (D_g^\top \Sigma^{-1} D_g + \sigma_a^{-2})^{-1}, \quad (46)$$

$$m_g = s_g^2 D_g^\top \Sigma^{-1} r. \quad (47)$$

The source evidence integrates a rather than inserting the true amplitude:

$$p(r | g) = \int p(r | a, g) p(a) da, \quad p(r | H_0) = \mathcal{N}(r; \mathbf{0}, \Sigma). \quad (48)$$

Normal versus event probability and the source posterior follow from these normalized densities and declared priors.

Finite changes bend the response. Symmetric development simulations estimate a source-specific second-order term Q_g and replace the event mean by

$$r_g(a) = aD_g + a^2 Q_g + O(a^3). \quad (49)$$

Four likelihood variants isolate mean curvature and amplitude-dependent covariance. DEV selects L2, which uses the second-order mean and fixed W2 covariance. Because that mean is quadratic in a , L2 evaluates (48) by normalized 401-point quadrature over $[-0.12, 0.12]$, with a mode-centered refinement for narrow posteriors; the analytic H_0 density uses the same evidence normalization. TEST then remains frozen. The descriptor-DAE tangent is incomplete, so (44) is numerical, not analytic.

This executed model conditions on the nominal initial state, assumes no corrupted channels, and uses a common fixed topology and known 30-frame onset window. It therefore

tests the amplitude-information and signature-coherence specialization in (36)–(37); it does not implement the nuisance-projected candidate manifolds in (26) or test Proposition 3 on the IEEE 39-bus system.

C. NOMINAL ESTIMATOR HIERARCHY

The estimator sequence changes one information source at a time. B0 retains only the equilibrium. B1 adds instantaneous PMU conditioning, B2 adds causal state propagation, and B3 tests retrospective temporal information. When the equilibrium becomes stale, E06-H estimates a physically admissible operating point through regularized nonlinear recentering. The full nonlinear Hybrid-DAE fixed-lag posterior is not evaluated in the experiments.

All nominal estimators use the same equilibrium y_0 , hidden-voltage center $z_{\mathcal{U},0}$, sampled dynamics A_d , PMU map C , and hidden-output map $L_{\mathcal{U}}$. This shared model isolates the value of measurement updating and temporal propagation.

The equilibrium baseline B0 returns

$$\hat{z}_{\mathcal{U},k}^{\text{B0}} = z_{\mathcal{U},0}. \quad (50)$$

It measures how much of the score can be obtained by remaining at the nominal power-flow solution.

The snapshot estimator B1 computes a prior-conditioned linear minimum-mean-square-error (LMMSE) update independently at every frame,

$$K_{\text{LMMSE}} = P_0 C^T (C P_0 C^T + R)^{-1}, \quad (51)$$

$$\hat{\delta x}_k^{\text{B1}} = K_{\text{LMMSE}}(y_k - y_0), \quad \hat{z}_{\mathcal{U},k}^{\text{B1}} = z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{\delta x}_k^{\text{B1}}. \quad (52)$$

B1 uses network-constrained cross-covariance but no state carried from frame $k - 1$.

Conditional on one linear-Gaussian candidate with no corruption, B2 is not an ad hoc module: the Kalman recursion computes the exact conditional mean and covariance in (39). Candidate uncertainty, nonlinear switching, unknown integrity support, and non-Gaussian discrepancy are what make the enclosing hybrid problem approximate.

The causal Kalman estimator B2 applies the classical linear-Gaussian recursion [55]:

$$\hat{x}_{k|k-1} = A_d \hat{x}_{k-1|k-1}, \quad (53)$$

$$P_{k|k-1} = A_d P_{k-1|k-1} A_d^T + Q, \quad (54)$$

$$\nu_k = (y_k - y_0) - C \hat{x}_{k|k-1}, \quad (55)$$

$$S_k = C P_{k|k-1} C^T + R, \quad (56)$$

$$K_k = P_{k|k-1} C^T S_k^{-1}, \quad (57)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \nu_k. \quad (58)$$

The covariance update uses the Joseph form. The hidden estimate is $z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{x}_{k|k}$. B2 is causal because only $y_{0:k}$ enters the result at k .

B3 is an exact fixed-lag Gaussian smoother, following the linear Gaussian backward-recursion principle in [56], evaluated at lags $\ell \in \{0, 1, 3, 5, 10, 15, 30, 60\}$ frames. It uses

measurements through $k + \ell$ to refine the estimate at k and is therefore retrospective by $\ell/30$ seconds. Dense and cached implementations agree numerically on representative lags. B3 tests whether temporal information beyond the causal update improves the hidden-voltage target; it is not used to support a real-time claim.

D. MARGINAL CALIBRATION AND COLORED DISCREPANCY

For the hidden two-component voltage error at bus i , B2 provides

$$\Sigma_{i,k} = L_i P_{k|k} L_i^T. \quad (59)$$

The original covariance is U_0 . A two-block temperature model U_2 is fitted on an independent calibration split:

$$\Sigma_{i,k}^{U_2} = T \Sigma_{i,k} T^T, \quad T = \text{diag}(\sqrt{\tau_{\Re}}, \sqrt{\tau_{\Im}}). \quad (60)$$

The temperatures minimize Gaussian negative log likelihood (NLL) on calibration data and are frozen before TEST. Coverage, NLL, normalized innovation squared (NIS), and temporal innovation correlation are reported separately; nominal marginal coverage alone is not called complete calibration.

The residual study considers a latent Gauss–Markov process

$$c_{k+1} = F_c c_k + \eta_k. \quad (61)$$

In D2-M, c_k enters the PMU measurement residual through a calibration-fitted low-rank basis. In D2-P, it enters a sparse physical process dictionary. Basis, rank, F_c , and latent covariance are selected using calibration records only. The experiment asks whether the observed low-rank structure improves reconstruction, hidden uncertainty, or innovation whiteness. It does not assume that a lower innovation autocorrelation is sufficient.

E. MODEL-ADEQUACY STATISTIC AND RECENTRING ABLATIONS

Let μ_ν and Σ_ν be the mean and shrunk covariance of B2 innovations on the nominal calibration set. The per-frame adequacy statistic is the five-frame trailing mean of

$$a_k = (\nu_k - \mu_\nu)^T \Sigma_\nu^{-1} (\nu_k - \mu_\nu). \quad (62)$$

Mismatch labels and hidden truth do not enter (62); hidden truth is used only after inference to label frames whose hidden HVRE exceeds the frozen evaluation threshold. AUROC and AUPRC therefore measure whether observed innovations anticipate unacceptable hidden error, not whether the detector identifies a mismatch family.

Three variants diagnose the failure mechanism without merging their evidence contracts. R1 replaces the nominal equilibrium with the true plant equilibrium and is explicitly oracle. R2-A estimates an eight-dimensional grouped observed-bus voltage offset using a 60-frame trailing window, nominal matrices, ridge regularization, and only PMU measurements. It is a causal negative ablation: its center/state cross-covariance is neglected and it does not enforce the nonlinear AC equations.

The corrected regularized physical estimator addresses M6 operating-point mismatch at one equilibrium snapshot. Let

$$d = \text{col}(\Delta P_{PQ}, \Delta Q_{PQ}, \Delta P_{PV}) \in \mathbb{R}^{43} \quad (63)$$

collect active and reactive changes at 17 admissible PQ loads and active-power changes at nine PV buses. Voltage is parameterized as $V(\theta, \rho) = \exp(\rho + j\theta)$. Defining the scaled PMU residual as $e_y(V) = D_y^{-1}[h_P(V) - y_P]$, the estimator solves

$$(\hat{V}, \hat{d}) = \arg \min_{V, d} \mathcal{J}(V, d), \quad (64)$$

$$\mathcal{J}(V, d) = \|e_y(V)\|_2^2 + \gamma^2 \|g_{AC}(V, d)\|_2^2 + \frac{\lambda}{q_d^2} \|d\|_2^2.$$

where h_P is the audited voltage/terminal-current operator, g_{AC} includes PQ balance, explicit PV-magnitude constraints, solved PV reactive power, and bus 31 as Slack, and D_y scales voltage and current components separately. The weights are $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$, selected on a disjoint DEV split; continuation initialization is then frozen for TEST. Neither the true nuisance vector nor the hidden voltages enter the TEST solve.

Because d may be non-unique, evaluation distinguishes nuisance information from target information. If J_{yd} and J_{zd} are the observed-PMU and hidden-voltage Jacobians with respect to d after eliminating voltage through the AC constraints, the local target residual is

$$r_{d \rightarrow z} = \frac{\|J_{zd} N(J_{yd})\|_F}{\max(\|J_{zd}\|_F, \epsilon)}, \quad (65)$$

where $N(J_{yd})$ spans the PMU-uninformed nuisance nullspace. Exact target identifiability would require a zero residual. The experiment instead tests the weaker empirical question: whether (64) reconstructs the hidden voltage accurately despite rank-deficient nuisance information, and whether its Laplace covariance is credible.

VI. VALIDATED BENCHMARK AND EXPERIMENTAL PROTOCOL

A. BENCHMARK PROVENANCE AND NUMERICAL GATES

The primary plant uses Julia 1.11.9, PowerDynamics.jl 5.0.0, and NetworkDynamics.jl 1.3.0 [57]. It contains 39 buses, 46 branches, 19 loads, 10 sixth-order Sauer–Pai machines, nine AVR Type-I controllers, and nine TGOV1 governors. The descriptor realization has 192 coordinates: 114 differential and 78 algebraic. These parameters are package-defined tutorial data. We therefore classify the plant as a numerically validated pedagogical benchmark rather than an authoritative representation of an external grid.

The validation ladder checks the model before estimator error is scored. A same-plant static translation was audited after correcting tap orientation, shunts, line charging, and full- Y_{bus} injection accounting. The largest remaining static discrepancy is 1.71×10^{-3} MVA in terminal flow. The controlled PMU operator agrees with the simulator to 2.71×10^{-7} pu in complex voltage and 1.43×10^{-14} pu in analytic/backend terminal current.

Dynamic validation checks the equilibrium, local modes, nonlinear trajectories, and linearization. The DAE equilibrium residual is 4.66×10^{-13} . After removing the reference mode, the least-damped physical pair is $-0.09818 \pm j0.42948 \text{ s}^{-1}$. Nonlinear Rodas5P simulations remain finite for the audited perturbations. Amplitude sweeps satisfy

$$\|y_{NL}(\epsilon) - y_{LIN}(\epsilon)\|_2 \propto \epsilon^p, \quad (66)$$

with fitted slopes between 1.982 and 2.000 for representative state, governor-reference, and load perturbations, consistent with first-order linearization error.

B. NOMINAL RECONSTRUCTION AND CALIBRATION

The nominal dataset contains 20 development and 100 test trajectories, each sampled at 30 Hz for 3 s. Perturbations are normal-operation changes to governor references, loads, or their combination. The equilibrium, linear model, $P_0 = 10^{-2}I$, $Q = 10^{-6}I$, and $R = 10^{-6}I$ are frozen before TEST. B0, B1, B2, and all B3 lags are evaluated on the same trajectories. A separate 30-trajectory calibration split fits U1/U2 temperatures and low-rank discrepancy variants; TEST is not used for selection.

Hidden truth is used only for post-hoc metrics. The eight observed buses are excluded from every primary reconstruction score. The phasor prediction is formed as $V_0 + \Delta V$, and angle error is wrapped using $\text{atan2}(\sin \Delta\theta, \cos \Delta\theta)$. These rules correct an earlier exploratory implementation that incorrectly interpreted $\angle \Delta V$ as the absolute angle; no superseded result from that implementation is retained.

C. PHYSICAL MISMATCH CAMPAIGN

For each mismatch case, a fresh PowerDynamics plant is rebuilt from copied source tables, followed by power flow, dynamic initialization, and nonlinear simulation. The original package CSVs remain hash-identical. The seven families are network $R/X/B$ (M1), machine parameters (M2), governor parameters (M3), AVR parameters (M4), ZIP/load model (M5), operating-point redispatch (M6), and coupled mismatch (M7). Stress scale is $m \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$. These are engineering stress-test ranges, not probability priors.

All 980 main-grid cases and all 175 refinement cases passed power-flow, initialization, and time-domain acceptance. B1, B2, U0, U2, and the adequacy score are frozen from nominal data. Bootstrap intervals use seeds as independent units. The plots read the frozen per-case table directly; a previously used refined breakpoint mixed aggregation definitions and is excluded.

D. ORACLE, OFFSET, AND PHYSICAL-MAP ADAPTATION CONTRACTS

The oracle subset compares the frozen estimator with true-equilibrium recentering and true local relinearization on the same mutated trajectories. Oracle quantities are evaluation-only. The offset-only recentering study uses new DEV seeds

TABLE 4. Frozen experimental contracts. State and event campaigns retain separate simulators, splits, targets, and denominators.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families, seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 candidate load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source–amplitude–noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; 9,600 weak + 800 no-event and 8,000 finite-amplitude TEST records	Source–amplitude–noise record
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied; exact source	303 decisions from 101 event parents $\times$ 3 seeds

101–110 and TEST seeds 201–220, disjoint from all earlier datasets. Its window, cadence, regularization, and activation threshold are selected on DEV and frozen before TEST. No event labels, hidden state, or mismatch family enter the R2-A estimate.

The corrected regularized physical-recentering campaign is a separate M6-only contract. It uses 30 fresh DEV cases from seeds 601–610 and 60 TEST cases from seeds 611–630, with 20 TEST equilibria at each of $m \in \{0.5, 1.0, 1.5\}$. DEV selects the nuisance scale, regularization, and continuation initialization from a small fixed grid. TEST uses only the observed voltage and terminal-current phasors; hidden voltages are revealed afterward for scoring. A solution is accepted only when the nonlinear optimizer succeeds, all outputs are finite, and the maximum AC residual is at most 5×10^{-5} . Rejections remain in the denominator; the frozen rejection table is empty. A six-case subset is also solved with an independent constrained optimizer, and its partial convergence status is reported rather than upgraded to full solver equivalence.

All numerical tables and empirical figures are regenerated from the hashed snapshot under `generated/frozen`. The build script checks the sample counts, acceptance flags, rank-deficient nuisance information, and preregistered adaptation-closure range before producing the tables and figures.

E. LOAD-SOURCE INFERENCE CONTRACT

The load-source campaign uses the PowerDynamics plant and the same eight PMU buses, but it has its own frozen normal banks, event dictionary, and test manifests. The dictionary covers 16 candidate load buses listed in its manifest. Centered $\pm 0.5\%$ nonlinear simulations define (44); the corresponding dictionary hash is fixed before the Bayesian tests. Candidate source and amplitude are inferred from the PMU window. The true source and true amplitude appear only in the evaluator.

V1 tests finite changes of ± 1 , ± 2.5 , ± 5 , and $\pm 7.5\%$ on 2,560 event and 320 no-event records. V2 expands normal calibration to 1,000 records, uses 500 normal and 640 event records for likelihood selection, then opens two TEST blocks. The weak block contains 9,600 events at magnitudes from 0.01 to 0.4% and 800 no-event records. The finite block contains 8,000 events from ± 1.25 to $\pm 9\%$. Noise seeds are disjoint across CAL, DEV, and TEST. A test record is one

source–amplitude–noise realization; repeated samples within its 30-frame window are never counted as independent cases.

W2 and L2 are selected on CAL/DEV, respectively, and frozen for both V2 TEST blocks. The weak-event operating threshold is 0.5 posterior event probability. No TEST row enters covariance estimation, likelihood selection, prior selection, or threshold tuning. The campaign covers one physical family and a fixed topology; it does not validate simultaneous events, missing PMUs, or cross-simulator transfer.

F. EVENT-DIAGNOSIS CAMPAIGN AND SIGNAL EVIDENCE

The separate ANDES campaign [58] uses the same eight PMU buses and 138 source configurations: one normal, 39 faults, 34 outages, ten generation, 18 load, eight missing-data, ten generation-and-missing, eight bad-data, and ten mixed configurations. Five replicas give 690 trajectories. Deterministic source/replica formulas vary severity over $\{0.06, 0.09, 0.12, 0.15\}$, operating scale over $\{0.97, 0.985, 1.00, 1.015, 1.03\}$, and the simulator/noise seed; source and 2-s onset stay fixed. Per-PMU AR(1) Gaussian noise is calibrated from normal, available RAW0001 rows. Trajectories last about 5 s; missing/corrupted intervals last 0.7 s. Physical interventions act in the simulator, while integrity changes act on sampled measurements.

Three replicas per source configuration enter training, one development, and one test, producing 414/138/138 complete-trajectory splits. Consequently, the primary event benchmark evaluates new replicas at sources already represented in training. The separate asset-disjoint audit removes the physical source from all source-labeled fitting and tuning. Results from these contracts are reported side by side but never interpreted as the same generalization task.

The historical baseline is a hierarchy of extremely randomized trees. Its 136-feature version uses trailing voltage magnitude, angle, frequency, missingness, and a normalized disturbance score. The 448-feature version adds multi-scale statistics, adaptive residuals, and simultaneous contrasts among PMUs. A deterministic availability gate handles directly observed dropout. These are physics-guided features and typed decision heads; they are not a DAE-constrained learning architecture, and the ensemble’s leaf-vote scores are not treated as calibrated probabilities.

Figure 5 uses the first labeled episode of six representative families in the 89.65-min RAW0001 reference simulation. The curves are measured simulator outputs, not sketches or smoothed reconstructions. They show why no single channel suffices: a fault creates a short voltage sag, an outage redistributes current, injection changes have slower frequency/current responses, and measurement failures require availability or phase-consistency information.

VII. NOMINAL RECONSTRUCTION AND STRUCTURAL EVIDENCE

A. WHAT THE SPARSE PMUS RECOVER

Table 5 gives the nominal TEST results. Conditioning the equilibrium estimate on the current PMU sample (B1) reduces hidden-voltage HVRE by more than one order of magnitude. The causal Kalman recursion (B2) improves on B1 in 69.0% of the same trajectories, but its paired mean advantage is only -0.000270 percentage points. Most of the nominal gain therefore comes from the measurement-conditioned physical map; temporal propagation supplies a smaller correction.

Figure 6(a) makes the trajectory distributions and scale separation visible, while panel (c) shows the paired differences rather than comparing overlapping marginal intervals. The equilibrium baseline is more than one order of magnitude worse than either measurement-updated estimator. This is not a comparison against learned or nonlinear DAE estimators; it is a controlled decomposition of equilibrium centering, instantaneous PMU conditioning, and causal temporal propagation within one local model.

B. FUNCTIONAL STRUCTURE AS A PREDICTOR OF ERROR

At horizon 180, the finite-horizon Gramian has pseudo-rank 39 of 114 and the aggregate target residual is nonzero; neither full-state nor exact functional observability is declared. The preregistered bus residual nevertheless correlates with B2 HVRE at Spearman $\rho = 0.5625$ [Fig. 6(b)], with a two-sided permutation p -value of 0.0012. Because the 31 buses share trajectories and network coupling, bus-resampling intervals are treated as exploratory rather than as independent-unit confidence intervals. The same qualitative relation appears for B1.

The relationship is informative but incomplete. Bus 36 has the largest structural residual, whereas the five largest B2 errors occur elsewhere. The projector captures one source of difficulty but not the nonlinear excitation distribution, estimator covariance, or every conditioning effect. Electrical distance to the nearest PMU has not been computed on the same audited network, so the analysis cannot yet show that $r_{F,i}$ adds information beyond that confounder. The residual remains a diagnostic, not a binary observability certificate.

C. RETROSPECTIVE SMOOTHING TEST

Across lags from one frame to 2 s, the implemented fixed-lag smoother never improves mean HVRE over lag zero; every positive lag is slightly worse in the nonlinear TEST trajectories [Fig. 7(a)]. A correctly specified linear-Gaussian

smoother cannot increase expected squared error, so this result diagnoses a violated modeling premise rather than a general failure of smoothing. The local model absorbs deterministic governor/load excitation into white process noise, while the lag-one innovation autocorrelation remains 0.602 and the Ljung–Box test rejects whiteness. An unknown-input estimator [12] and a re-estimated process model are required before any latency–accuracy conclusion is drawn.

VIII. UNCERTAINTY, PHYSICAL MISMATCH, AND ADAPTATION

A. MARGINAL CALIBRATION DOES NOT WHITEN INNOVATIONS

The nominal B2 covariance is conservative and covers essentially all real and imaginary errors at the nominal 95% level. Two temperatures fitted on 30 independent CAL trajectories bring the TEST marginal coverage closer to its target and improve hidden NLL (Table 6). They do not change the lag-one innovation autocorrelation, and the Ljung–Box test still rejects whiteness. Marginal calibration and temporal model adequacy are therefore separate properties [Fig. 7(b)–(c)].

Four principal components explain 99.2% of the calibration-residual variance, but low rank alone does not make the residual a useful process model. A rank-four measurement Gauss–Markov correction lowers the lag-one autocorrelation, then worsens HVRE and undercovers both voltage components. Placing one mode in the process model changes neither HVRE nor autocorrelation appreciably. Colored residual structure must therefore be tied to a physical discrepancy mechanism before it improves hidden-state uncertainty.

B. MISMATCH RESPONSE DEPENDS ON MODEL FAMILY

The rebuilt-plant campaign reveals a clear family dependence (Fig. 8 and Table 7). At the largest tested scale, network mismatch raises the median error by $16.99\times$ and an operating-point shift raises it by $8.32\times$. Machine, governor, AVR, and load-model perturbations remain near the nominal median because this 3-s excitation does not measurably expose those parameter changes. Their near-1 $\times$ ratios are regime-limited null responses, not evidence of robustness. The tested estimator fails visibly when the algebraic network map or equilibrium center changes.

Nominal covariance repair does not transfer automatically. Averaged across all mismatch cases, U0 has 92.9% 95% coverage and nominally fitted U2 has only 47.5%. The reduction reflects physical shift, not a calibration/test leak: U2 remains frozen from nominal CAL data. This result limits the interpretation of Section V: nominal dispersion scaling is not robustness to a changed plant.

C. ADEQUACY DETECTION AND PHYSICAL RECENTERING

The observed-PMU innovation statistic detects many cases in which hidden HVRE is unacceptable. At the largest tested scale, its macro AUROC/AUPRC is 0.928/0.787 [Fig. 8(c)]. Family-level AUROC is omitted when every frame has the

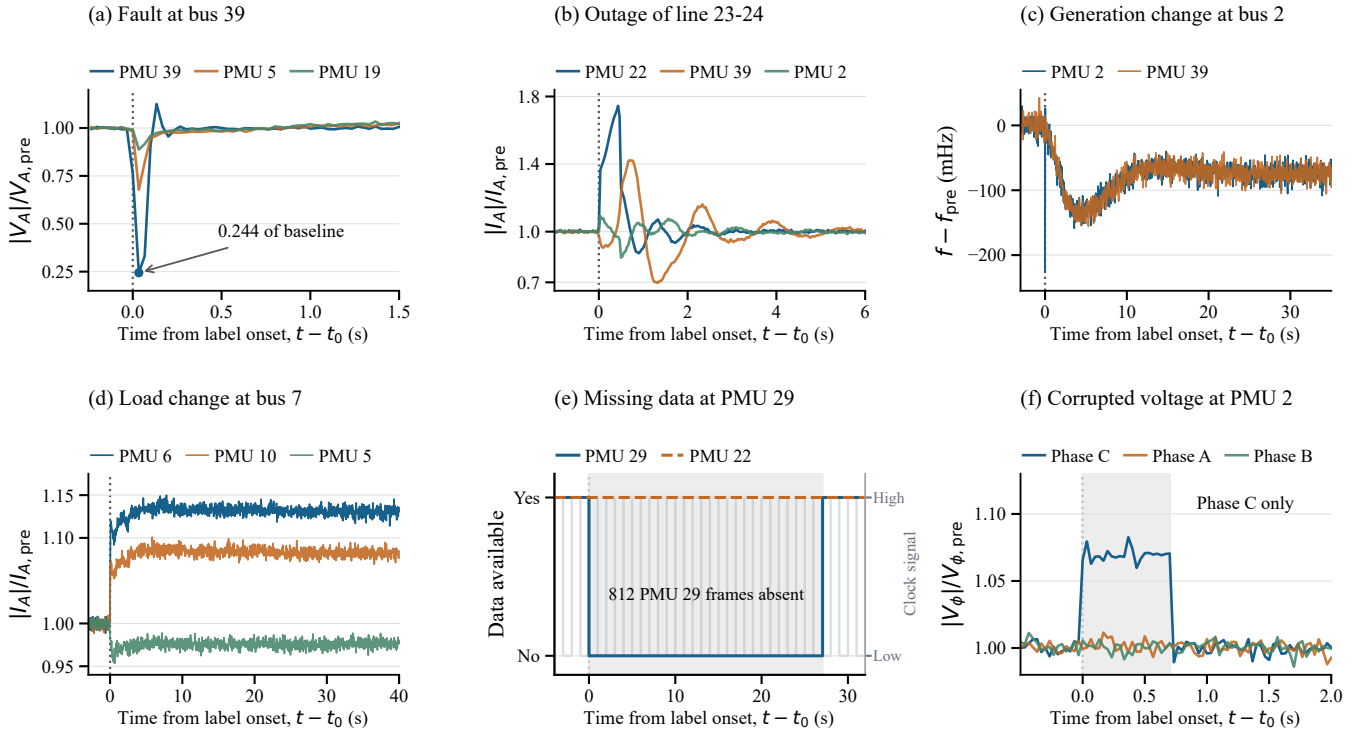


FIGURE 5. Representative measured event signatures from RAW0001, sampled at approximately 30 frames/s. (a) Bus fault, (b) line outage, (c) generation change, (d) load change, (e) missing data, and (f) single-phase corruption. The 812-frame missing interval in (e) is approximately 27.1 s; the full-height light-gray square wave and right-hand axis show a schematic reference clock derived from the recorded time coordinate, not a received PMU channel. Thus time advances while PMU 29 packets are absent. Dotted lines mark the first labeled sample; gray shading marks missing or corrupted intervals. Magnitudes are normalized by the preceding 3-s median, frequency is relative to its corresponding median, and time scales differ by panel. The plotted measurements and source-file hashes are frozen in the evidence snapshot.

TABLE 5. Nominal reconstruction on 100 frozen nonlinear TEST trajectories. HVRE is the trajectory-and-bus mean relative complex-voltage error over the 31 unobserved buses; intervals bootstrap independent trajectories.

Method	Complex RMSE	$ V $ RMSE	Angle RMSE ($^\circ$)	Mean HVRE (%)	95% HVRE interval (%)
B0 equilibrium	3.486e-03	2.496e-05	0.19497	0.340200	[0.293231, 0.387428]
B1 snapshot LMMSE	1.529e-04	5.091e-05	0.00821	0.009904	[0.008342, 0.011514]
B2 causal Kalman	1.338e-04	4.235e-05	0.00722	0.009634	[0.008006, 0.011302]

TABLE 6. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

same frozen adequacy label; AUPRC remains defined. The statistic can flag a stale model, but it neither identifies the mismatch family nor supplies a calibrated alarm probability.

The earlier R2-A offset-only ablation remains scientifically useful because it isolates what does not solve the problem. Across nonnominal identifiable M1, M6, and M7 cases it closes only 0.4% of the oracle gap, despite passing its nominal-safety gate. That failure does not imply that adaptation from observed PMUs is impossible; it shows that grouped voltage offsets without AC balance and a rebuilt physical

center are inadequate.

The regularized-recentering experiment tests that distinction for M6. Every TEST case satisfies the prespecified convergence gate. Across the three mismatch scales, the physical solve removes 97.4%–98.3% of the error gap to an oracle center [Fig. 9(a)–(b)]. The median solve takes 61.7 ms. Table 8 retains the point errors, paired intervals, and tail runtime; these static timings do not establish end-to-end real-time operation.

The nuisance information matrix has median rank 25

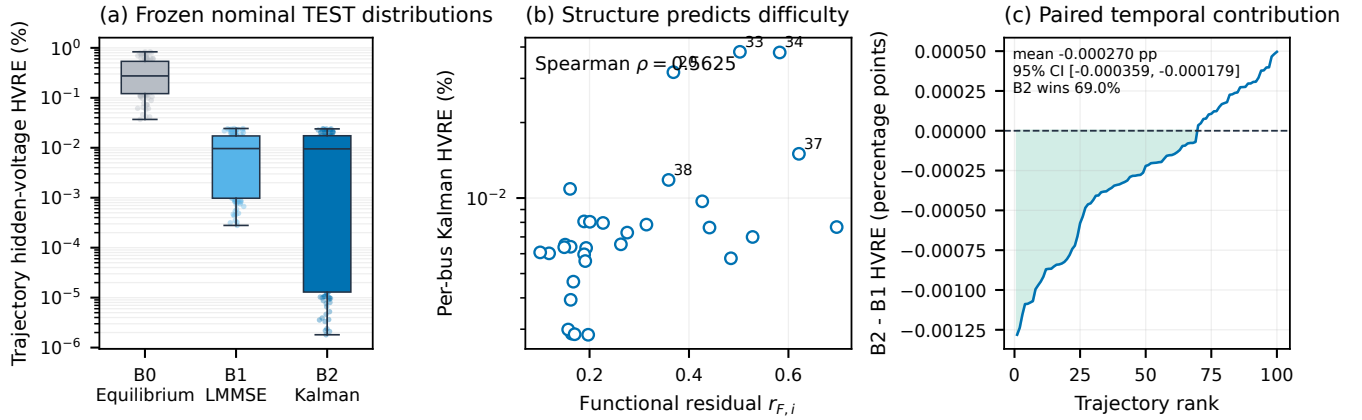


FIGURE 6. Nominal reconstruction and structural evidence. (a) Trajectory-level hidden-voltage HVRE distributions on the same 100 nonlinear TEST trajectories. (b) Per-bus causal Kalman HVRE versus the horizon-180 functional residual exported before TEST scoring; labels identify the five buses with largest empirical error. (c) Sorted paired B2-minus-B1 HVRE differences; negative values favor the causal estimator, and the annotation reports the paired bootstrap interval and win rate.

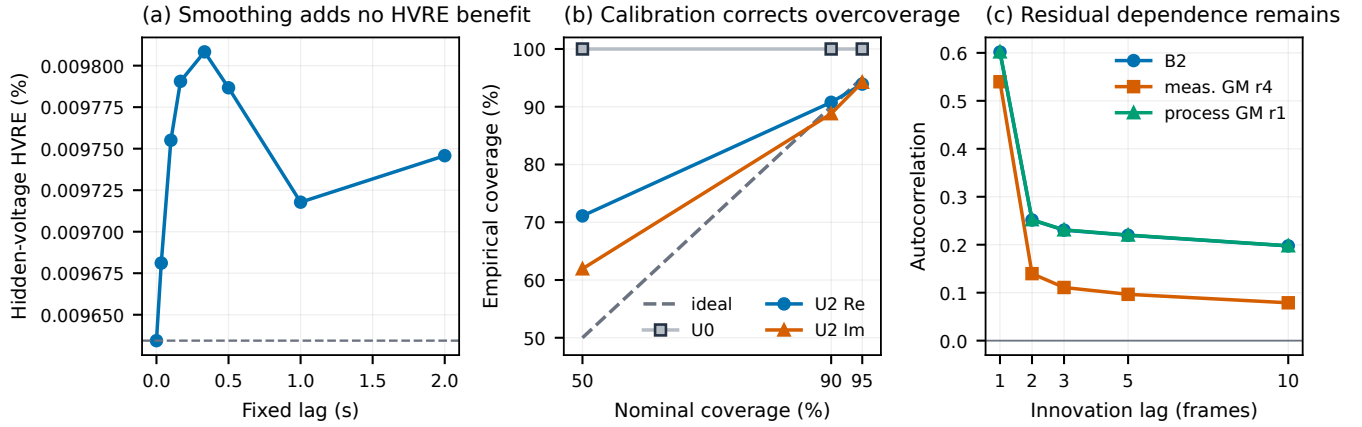


FIGURE 7. Temporal and uncertainty diagnostics on the nominal TEST set. (a) Exact fixed-lag smoothing; lag zero is the causal Kalman result. (b) Marginal real-component coverage for U0 and real/imaginary U2 blocks against nominal coverage. (c) Innovation autocorrelation for the original estimator and calibration-selected low-rank discrepancy placements.

TABLE 7. Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

among 43 coordinates. Even with that nullspace, the requested voltage field is recovered with two orders of magnitude less error than the frozen center. This supports target recovery despite nuisance non-uniqueness. It does not prove exact functional identifiability because the local hidden-voltage residual remains nonzero, nor does it show that the injection vector is recovered.

Point accuracy does not certify posterior uncertainty. The

pseudoinverse Laplace covariance is formed only on locally informed nuisance directions; because $J_{zd}N(J_{yd}) \neq \mathbf{0}$ in (65), it does not represent the image of the PMU-uninformed nuisance nullspace in the hidden-voltage target. This explains how marginal intervals can cover every component while the joint NEES-like statistic grows with mismatch and remains far above its nominal reference [Fig. 9(c)]. The independent constrained solver also converges on only part of its small diagnostic subset. M6 point reconstruction is supported; joint uncertainty, network mismatch, coupled mismatch, and sequential adaptation are not.

D. PHYSICS-BASED LOAD DETECTION AND SOURCE RESOLUTION

The centered numerical tangent passes two checks before source inference is scored. Across all 16 load candidates, its largest discrepancy from either one-sided derivative is 0.235%, and the minimum cosine between the positive and negative perturbation directions is 0.999995. At a separate 10% stress, the median and maximum relative trajectory er-

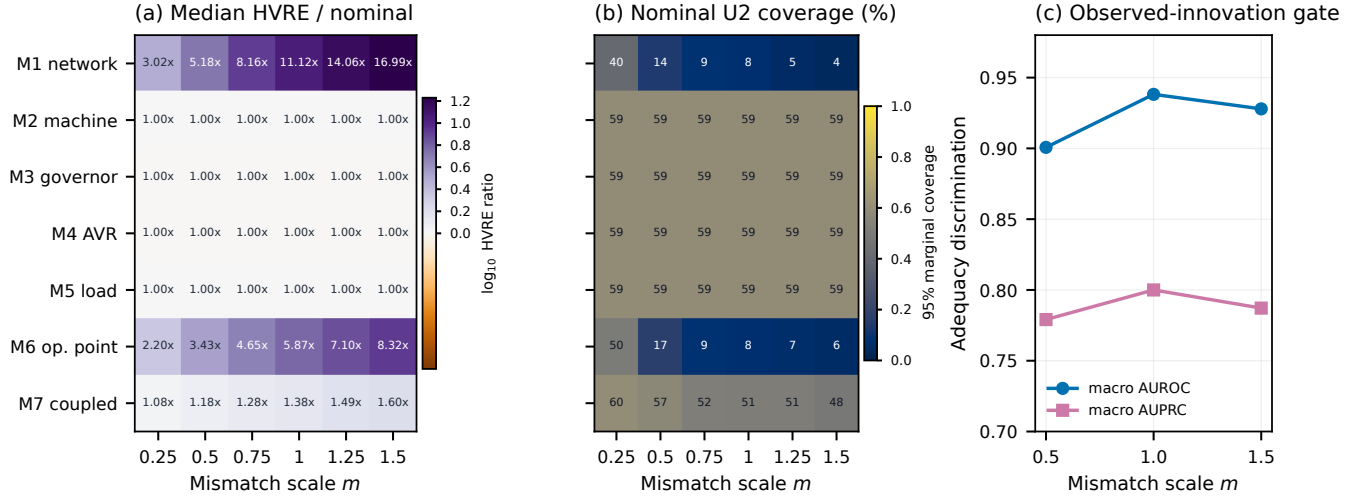


FIGURE 8. Physical-mismatch atlas for the fixed nominal estimator. (a) Median B2 hidden-voltage HVRE relative to the pooled $m = 0$ median for every family-scale cell; annotations show the untransformed ratio and color encodes its base-10 logarithm. (b) Mean nominal-U2 marginal 95% coverage under the same rebuilt plants. (c) Macro adequacy discrimination from observed innovations; undefined family-level AUROCs are omitted from the macro mean. Each family-scale cell contains 20 independently rebuilt plants.

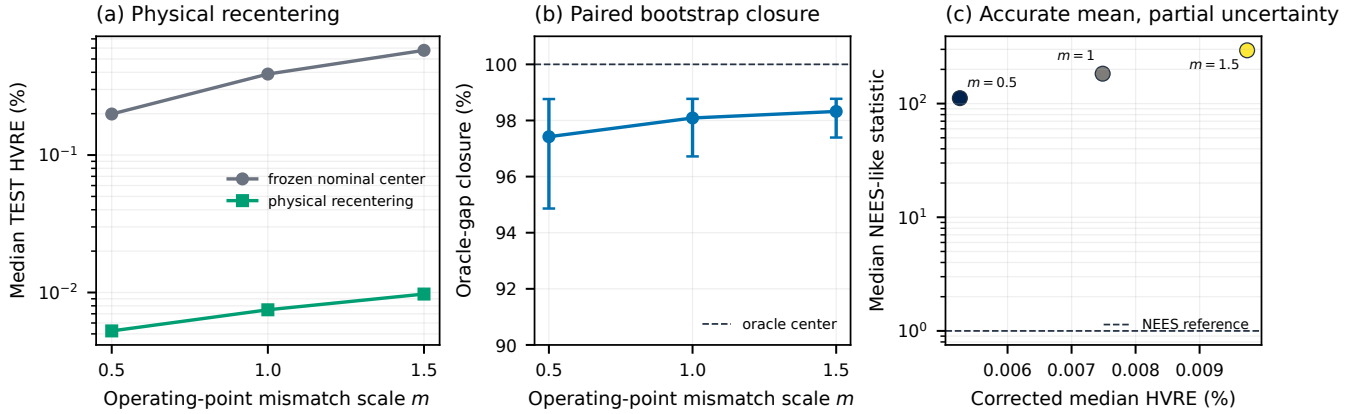


FIGURE 9. Corrected regularized physical recentering for held-out M6 operating-point mismatch. (a) Median hidden-voltage HVRE for the nominal center and the physical solve. (b) Paired oracle-gap closure with 95% bootstrap intervals over 20 independent TEST cases per scale; the oracle-center reference is 100%. (c) Corrected point error versus the median joint NEES-like diagnostic. Marginal 95% coverage is 100% at each scale, but the joint statistic remains far above its nominal reference.

rors of the tangent prediction are 1.56% and 4.55%. These tests support the numerical physical dictionary over the stated range; they do not complete the pending analytic descriptor-DAE derivative.

Normal-data covariance changes the decision rule materially. The deliberately unscaled W0 identity diagnostic produces a 100% false-positive rate and 10.9% source Top-1; it is not an honest competitive covariance baseline. Channel covariance plus temporal AR(1) whitening gives 0.625% false positives and 100.0% Top-1 on 2560 event records. The larger V2 normal bank confirms the statistical mechanism: W2 has NIS per frame 0.998, lag-one autocorrelation -0.0026, and Ljung-Box $p = 0.598$. Channel covariance alone leaves lag-one autocorrelation at 0.341. A CAL-fitted scaled-identity baseline remains to be run.

The nonlinear remainder follows the expected local order. A log-log fit over candidate buses and development

amplitudes gives $p = 2.00$ with bootstrap 95% interval [1.79, 2.24] [Fig. 10(a)]. The second-order mean model L2 is then selected on DEV with 95.0% amplitude coverage. On the independent finite-amplitude TEST block, its 95% coverage ranges from 92.9% to 96.0% over signed changes from 1.25 to 9% [Fig. 10(f)].

Weak events expose the difference between detection and localization. At $|a| = 0.01\%$, sign-pooled AUROC is 0.733 while exact source Top-1 is 13.0%; at $|a| = 0.05\%$, the same quantities are 0.995 and 73.4%. Source Top-1 reaches 100% at $|a| = 0.4\%$ (Table 9). The event false-positive rate is 0.25% on 800 no-event records. Thus a window can contain detectable event evidence while remaining insufficient for a reliable source decision.

Amplitude Fisher information is negatively associated with the empirical amplitude required for 90% detection, with Spearman $\rho = -0.794$ over 16 sources [Fig. 10(e)]. This

TABLE 8. Corrected physical static recentering on 60 held-out M6 operating-point cases. Gap-closure intervals bootstrap the 20 independent TEST cases at each scale. Rank and nullity refer to the 43-dimensional nuisance basis; runtime is per solve.

m	n	Frozen HVRE (%)	Corrected HVRE (%)	Gap closure [95% CI] (%)	Rank	Nullity	Median NEES	Runtime (ms)
0.5	20	0.19885	0.00525	97.4 [94.9, 98.8]	25/43	18	111.6	57.6
1.0	20	0.38834	0.00749	98.1 [96.7, 98.8]	25/43	18	183.3	61.1
1.5	20	0.57795	0.00974	98.3 [97.4, 98.8]	25/43	18	294.2	69.2

TABLE 9. Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.

$ a $ (%)	AUROC	FPR (%)	FNR (%)	Top-1 (%)	Top-3 (%)
0.010	0.733	0.25	83.1	13.0	37.1
0.020	0.906	0.25	45.0	31.8	69.0
0.050	0.995	0.25	4.5	73.4	95.9
0.100	1.000	0.25	0.2	93.2	99.4
0.200	1.000	0.25	0.0	98.9	100.0
0.400	1.000	0.25	0.0	100.0	100.0

is exploratory: each 90% threshold is selected from only six tested magnitudes, so the result has visible discretization and no fitted psychometric interval or permutation p -value. The profiled pairwise information has $\rho = 0.361$ against a pooled 16-way confusion statistic. That comparison mismatches a pairwise quantity to an amplitude-marginalized multicandidate outcome and is not treated as a validation test. A proper bridge requires predicted versus empirical Top-1 for each source–amplitude pair.

E. KNOWN-SOURCE DETECTION AND LOCALIZATION

On the known-source TEST split, the availability-gated hierarchy reaches detection F1 0.989 but only 0.592 physical Top-1 (Table 10). Expanding the representation adds 9.6 percentage points to source ranking over the compact hierarchy. Multiscale dynamics, adaptive residuals, and PMU contrasts therefore help when the source is represented during training; the availability rule contributes mainly to integrity handling.

The gap between detection F1 and physical Top-1 confirms that a recognizable disturbance need not have a uniquely recoverable source. False alarms decrease to 1.45 episodes per normal-labeled minute, but the normal exposure is only about 7.15 min per split. That rate is evidence of alarm burden in a short synthetic stream, not an operational false-alarm guarantee.

F. SOURCE HOLDOUT BREAKS LEARNED LOCALIZATION

With the physical source excluded from source-labeled fitting and the true event family supplied, only 3 of 303 decisions are exact. Detection and event-family confusion have been removed from this test. The remaining error shows that the learned location heads and candidate descriptors retain source-specific associations, invalidating a zero-shot-source claim for the legacy system.

Transfer to RAW0001 is also weak: the basic candidate model localizes only 0.3 of five physical episodes on average. The simulator bank omits some channels used by the longer-

window implementation, so no single feature family can be blamed for the deficit. Internal simulation accuracy does not establish cross-record reliability.

Supervised, physics-guided features work for detection and known-source diagnosis but not for source-generalizable inference. The physical load experiment supplies a contrasting result: candidate sources are evaluated through simulated responses rather than learned asset labels. Its 16-source success is restricted to one event family, fixed topology, and matched simulator; it does not repair the broader held-source or cross-record failure.

IX. DISCUSSION AND LIMITATIONS

The three contracts expose different parts of the inverse problem. Eight voltage/current PMUs recover the requested 31-bus voltage field accurately in the PowerDynamics state tests even though the 114-state local model is not fully observable. A separate physical load-response dictionary supports normalized event and source evidence on the same simulator. The ANDES bank measures what the earlier supervised representation retained when source labels were removed. Low hidden-voltage HVRE still does not guarantee source separation, and high event visibility does not guarantee exact localization.

The local theory explains this split. Proposition 1 removes unobservable directions that do not affect the requested voltage function, and Corollary 1 attaches a covariance to that estimable function. Source diagnosis has a different geometry. Proposition 2 identifies intersecting candidate–nuisance response sets. Proposition 3 then prevents a stronger but invalid inference: zero first-order separation need not imply nonlinear impossibility because relative curvature can leave second-order evidence. Conversely, a positive affine margin is certified only while the nonlinear remainder stays below it. Large faults, topology switches, and unmodeled controls may violate both local descriptions.

Two theory-to-data links receive preliminary tests. The functional residual correlates with bus-level voltage error, but its projector fails rank monotonicity and needs an SVD/noise-tied recomputation; excitation, covariance, and electrical distance also confound it. Amplitude Fisher information is associated with the source-wise detection threshold, with the expected sign. Six threshold magnitudes over 16 sources make this exploratory. The pairwise Fisher quantity was compared with pooled 16-way confusion, a mismatched test rather than evidence against the pairwise theory. The new relative-curvature result is verified only by an algebraic toy check;

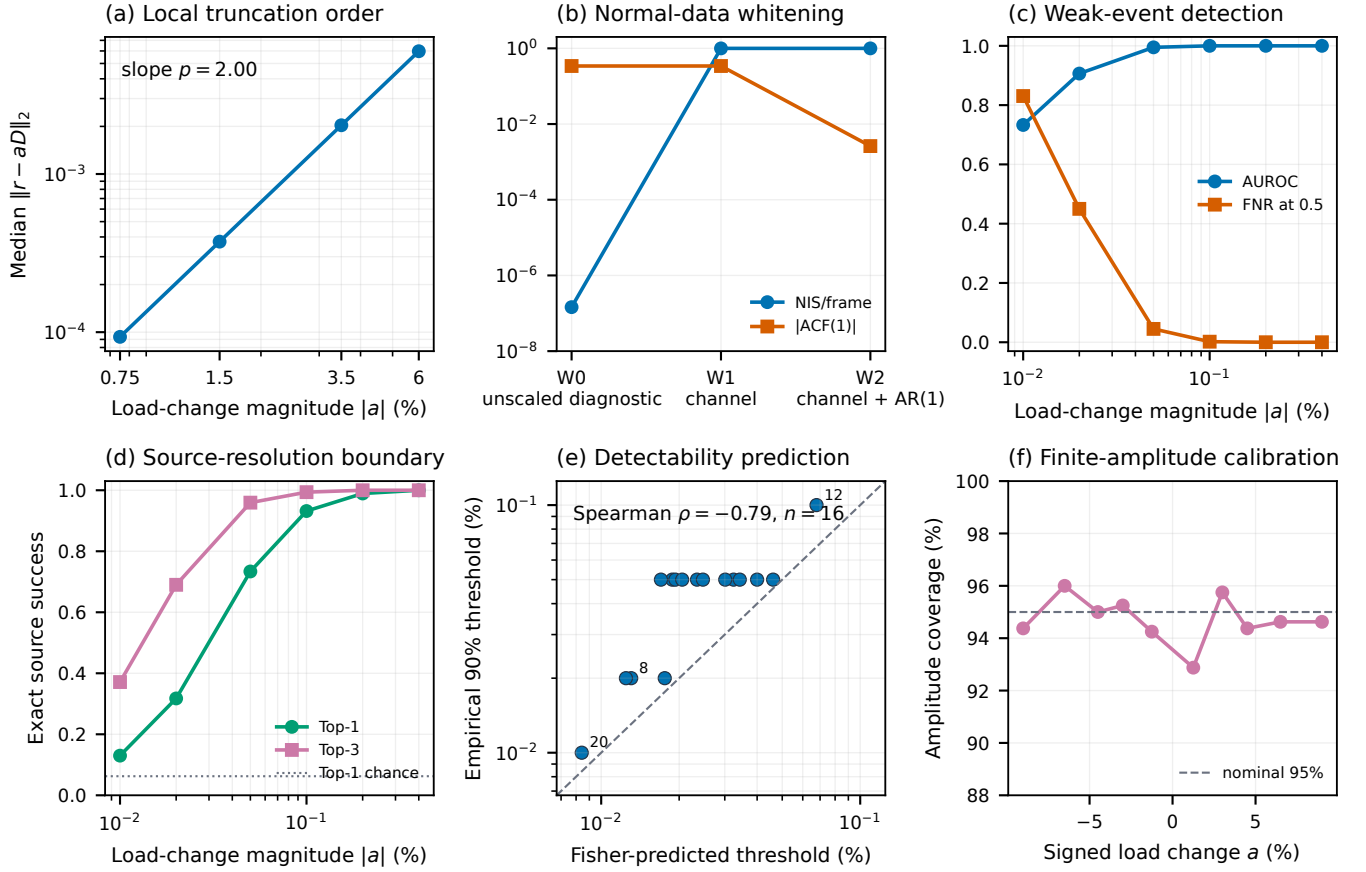


FIGURE 10. Frozen physics-based load-event evidence. (a) First-order residual scaling over 16 candidate buses. (b) Normal-data covariance diagnostics; W0 is an unscaled diagnostic, not a competitive baseline. (c) Weak-event detection and (d) exact source resolution, pooled only across signs at the same magnitude. (e) Fisher-information threshold against the discretized empirical 90%-detection amplitude ($n = 16$ sources); labels identify representative buses. (f) L2 amplitude coverage on the separate finite-amplitude TEST block. The weak block contains 9600 event and 800 no-event records; the finite block contains 8000 events.

TABLE 10. Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

evaluating it on IEEE 39-bus candidates requires separately estimated nuisance tangent and curvature.

The operating-point study also separates target recovery from parameter recovery. Regularized physical recentering closes 98.1% of the oracle-center gap on M6 although only 25 of 43 nuisance directions are locally informed. The hidden-voltage target can therefore be accurate while the injection coordinates remain non-unique. This result is confined to an equilibrium snapshot. It does not cover line outages, faults, or sequential event-conditioned estimation, and the local curvature covariance remains under-dispersed in the joint NEES-like statistic.

The legacy event model supplies one reusable mechanism and one warning. Multiscale summaries, adaptive residuals, and simultaneous PMU contrasts improve known-source

physical ranking by 9.6 percentage points. Availability belongs in the integrity state because a missing measurement is observed directly. Source identities and endpoint descriptors, however, allow memorization. Exact localization falls to 1.0% under complete physical-source holdout. The load experiment avoids an asset-label head by evaluating each candidate as a physical intervention. Learned residual features may still help, but their contribution must be measured against this physics-only likelihood.

The principal evidence gap is a multi-family end-to-end experiment. The three studies have different targets or simulator semantics and are simulated. The Bayesian result covers one family, fixed topology, known onset, closed candidates, matched operation, and synthetic noise; it does not fit state/integrity nuisance. The ANDES records have fixed

onsets and about 7.15 min of normal exposure per split. Neither field validity nor operational false-alarm control is established.

A decisive follow-up compares learned-only, physics-only, learned-residual, and diagnosability-aware methods under one audited plant and PMU operator. Event parents and sources must remain disjoint across fitting, calibration, and test. Frozen shift, topology, and PMU-availability regimes plus long normal exposure must report joint correctness, candidate-set coverage, delay, false alarms per hour, HVRE, calibration, runtime, and failed solves. Until then, multi-family superiority remains unproved.

X. CONCLUSION

Sparse-PMU monitoring contains two inverse problems with different information limits. Functional observability determines whether the requested hidden-voltage function is recoverable and gives the covariance of its minimum-variance linear estimate. Candidate-manifold separation determines whether two event sources remain distinguishable after state and integrity nuisance are fitted. When nuisance absorbs the event tangent, relative curvature can preserve second-order local evidence; when a positive affine margin exists, a bounded DAE remainder determines whether it survives nonlinearity.

The experiments show the distinction. The causal Power-Dynamics estimator attains 0.009634% mean hidden-voltage HVRE, and physical recentering removes 97.4%–98.3% of the operating-point error gap. In the executed 16-source load case, amplitude Fisher information is associated with a discretized 90%-detection threshold, while source Top-1 falls from 100% at 0.4% magnitude to 13.0% at 0.01%. The separate ANDES model detects known-source events, yet complete physical-source holdout yields only 3 correct decisions among 303.

Physical candidate responses now have direct evidence for one matched event family. Extending them to faults, outages, generation changes, integrity events, changing PMU sets, and simulator shift remains the next source-disjoint test. The central consequence is narrower: state recoverability, event detectability, and source diagnosability must be measured separately before any joint decision is trusted.

REFERENCES

- [1] A. G. Phadke and J. S. Thorp, *Synchronized Phasor Measurements and Their Applications*. New York, NY, USA: Springer, 2008.
- [2] J. Zhao, J. Qi, Z. Huang, A. P. S. Meliopoulos, A. Gómez-Expósito, M. Netto *et al.*, "Power system dynamic state estimation: Motivations, definitions, methodologies and future work," *IEEE Transactions on Power Systems*, vol. 34, no. 4, pp. 3188–3198, 2019.
- [3] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.
- [4] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.
- [5] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.
- [6] M. Rimkus, P. Kokoszka, D. Duan, X. Wang, and H. Wang, "Graph neural networks for the localization of faults in a partially observed regional transmission system," *Scandinavian Journal of Statistics*, vol. 52, no. 2, pp. 572–594, 2025.
- [7] Z.-M. Zhai, M. Moradi, and Y.-C. Lai, "Detecting attacks and estimating states of power grids from partial observations with machine learning," *PRX Energy*, vol. 4, no. 1, p. 013003, 2025.
- [8] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.
- [9] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.
- [10] J. Zhao and L. Mili, "A robust generalized-maximum likelihood unscented kalman filter for power system dynamic state estimation," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 4, pp. 578–592, 2018.
- [11] Z. Li, S. Li, B. Liu, S. S. Yu, and P. Shi, "A stochastic event-triggered robust cubature kalman filtering approach to power system dynamic state estimation with non-gaussian measurement noises," *IEEE Transactions on Control Systems Technology*, vol. 31, no. 2, pp. 889–896, 2023.
- [12] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.
- [13] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.
- [14] B. Qu, Z. Wang, B. Shen, H. Dong, and H. Liu, "Event-based joint state and unknown input estimation for energy networks: Handling multi-machine power grids," *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 1, pp. 253–264, 2023.
- [15] B. L. H. Nguyen, T. V. Vu, J. M. Guerrero, M. Steurer, K. Schoder, and T. Ngo, "Distributed dynamic state-input estimation for power networks of microgrids and active distribution systems with unknown inputs," *Electric Power Systems Research*, vol. 201, p. 107510, 2021.
- [16] Q.-H. Ngo, B. L. H. Nguyen, T. V. Vu, J. Zhang, and T. Ngo, "Physics-informed graphical neural network for power system state estimation," *Applied Energy*, vol. 358, p. 122602, 2024.
- [17] H. Yue, W. Zhang, U. C. Yilmaz, T. Yildiz, H. Huang, H. Liu, Y. Lin, and A. Abur, "Graph-learning-assisted state estimation using sparse heterogeneous measurements," *Electric Power Systems Research*, vol. 235, p. 110644, 2024.
- [18] S. Moshtagh, B. Azimian, M. Golgol, and A. Pal, "Topology-aware graph neural network-based state estimation for PMU-unobservable power systems," *IEEE Transactions on Power Systems*, 2025.
- [19] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.
- [20] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 6, pp. 1140–1153, 2014.
- [21] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.
- [22] M. Hou and R. J. Patton, "Input observability and input reconstruction," *Automatica*, vol. 34, no. 6, pp. 789–794, 1998.
- [23] F. Pasqualetti, F. Dörfler, and F. Bullo, "Attack detection and identification in cyber-physical systems," *IEEE Transactions on Automatic Control*, vol. 58, no. 11, pp. 2715–2729, 2013.
- [24] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.
- [25] W. Li, M. Wang, and J. H. Chow, "Real-time event identification through low-dimensional subspace characterization of high-dimensional synchrophasor data," *IEEE Transactions on Power Systems*, vol. 33, no. 5, pp. 4937–4947, 2018.
- [26] Y. Yuan, Y. Guo, K. Dehghanpour, Z. Wang, and Y. Wang, "Learning-based real-time event identification using rich real PMU data," *IEEE Transactions on Power Systems*, vol. 36, no. 6, pp. 5044–5055, 2021.
- [27] H. Li, Z. Ma, and Y. Weng, "A transfer learning framework for power system event identification," *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4424–4435, 2022.

- [28] Y. Yuan, Y. Wang, and Z. Wang, "Learning latent interactions for event classification via graph neural networks and PMU data," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 617–629, 2023.
- [29] Y. Cheng, N. Yu, B. Foggo, and K. Yamashita, "Online power system event detection via bidirectional generative adversarial networks," *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4807–4818, 2022.
- [30] A. A. Hai, T. Dokic, M. Pavlovski, T. Mohamed, D. Saranovic, M. Alqudah, M. Kezunovic, and Z. Obradovic, "Transfer learning for event detection from PMU measurements with scarce labels," *IEEE Access*, vol. 9, pp. 127 420–127 432, 2021.
- [31] W. Li, D. Deka, M. Chertkov, and M. Wang, "Real-time faulted line localization and PMU placement in power systems through convolutional neural networks," *IEEE Transactions on Power Systems*, vol. 34, no. 6, pp. 4640–4651, 2019.
- [32] S. Azizi, M. R. R. Jegarlupei, A. S. Dobakhshari, G. Liu, and V. Terzija, "Wide-area identification of the size and location of loss of generation events by sparse PMUs," *IEEE Transactions on Power Delivery*, vol. 36, no. 4, pp. 2397–2407, 2021.
- [33] S. Azizi, M. R. R. Jegarlupei, A. M. Kettner, and A. S. Dobakhshari, "Wide-area line outage monitoring by sparse phasor measurements," *IEEE Transactions on Power Systems*, vol. 38, no. 4, pp. 3621–3631, 2023.
- [34] Z. Wang, M. Jiao, D. Wang, M. Liu, M. Jiang, H. Wang, and S. Li, "A disturbance localization method for power system based on group sparse representation and entropy weight method," *Energy Engineering*, vol. 121, no. 8, pp. 2275–2291, 2024.
- [35] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
- [36] N. Rovatsos, Y. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.
- [37] A. Kummerow, C. Monsalve, and P. Bretschneider, "Siamese recurrent neural networks for the robust classification of grid disturbances in transmission power systems considering unknown events," *IET Smart Grid*, vol. 5, no. 1, pp. 51–61, 2022.
- [38] A. Kummerow, M. Dirbas, C. Monsalve, and P. Bretschneider, "Siamese sigmoid networks for the open classification of grid disturbances in power transmission systems," *IET Smart Grid*, vol. 6, no. 2, pp. 136–146, 2023.
- [39] Y. Hu and Z. Cheng, "Generative AI for open-set event detection and classification in power systems using synchrophasor data," *IEEE Transactions on Industrial Informatics*, 2026, early access.
- [40] B. Foggo, K. Yamashita, and N. Yu, "pmuBAGE: The benchmarking assortment of generated PMU data for power system events," *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 3485–3496, 2024.
- [41] K. Chatterjee, K. Mahapatra, and N. R. Chaudhuri, "Robust recovery of PMU signals with outlier characterization and stochastic subspace selection," *IEEE Transactions on Smart Grid*, vol. 11, no. 4, pp. 3346–3358, 2020.
- [42] S. Biswas, T. Yin, S. A. R. Naqvi, J. Follum, A. C. Varghese, T. Ahmad, and P. V. Etingov, "An open-access repository of synchrophasor data quality examples: Curation and example applications," *IEEE Access*, vol. 13, pp. 66 445–66 457, 2025.
- [43] A. Kummerow and P. Bretschneider, "Influence of nuisance variables on the PMU-based disturbance classification in power transmission systems," *at-Automatisierungstechnik*, vol. 71, no. 10, pp. 867–877, 2023.
- [44] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.
- [45] T. Zhang, T. L. Fernando, and M. Darouach, "Functional observability, functional detectability, and structural functional observability of linear systems," *IEEE Transactions on Automatic Control*, vol. 70, no. 3, pp. 1592–1607, 2025.
- [46] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.
- [47] M. Nyberg, "Criteria for detectability and strong detectability of faults in linear systems," *International Journal of Control*, vol. 75, no. 7, pp. 490–501, 2002.
- [48] M. Showkatbakhsh, Y. Shoukry, S. N. Diggavi, and P. Tabuada, "Securing state reconstruction under sensor and actuator attacks: Theory and design," *Automatica*, vol. 116, p. 108920, 2020.
- [49] Q. Sun and G.-H. Yang, "Joint estimation of states, sparse sensor attacks, and unknown inputs in discrete-time cyber-physical systems," *IEEE Transactions on Cybernetics*, vol. 56, no. 8, pp. 4625–4633, 2026.
- [50] A. Rotnitzky, D. R. Cox, M. Bottai, and J. M. Robins, "Likelihood-based inference with singular information matrix," *Bernoulli*, vol. 6, no. 2, pp. 243–284, 2000.
- [51] J. Qi, K. Sun, and W. Kang, "Optimal PMU placement for power system dynamic state estimation by using empirical observability gramian," *IEEE Transactions on Power Systems*, vol. 30, no. 4, pp. 2041–2054, 2015.
- [52] K. Sun, J. Qi, and W. Kang, "Power system observability and dynamic state estimation for stability monitoring using synchrophasor measurements," *Control Engineering Practice*, vol. 53, pp. 160–172, 2016.
- [53] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.
- [54] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.
- [55] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.
- [56] H. E. Rauch, F. Tung, and C. T. Striebel, "Maximum likelihood estimates of linear dynamic systems," *AIAA Journal*, vol. 3, no. 8, pp. 1445–1450, 1965.
- [57] A. Plietzsch, R. Kogler, S. Auer, J. Merino, A. G. de Muro, J. Liße, C. Vogel, and F. Hellmann, "PowerDynamics.jl—an experimentally validated open-source package for the dynamical analysis of power grids," *SoftwareX*, vol. 17, p. 100861, 2022.
- [58] H. Cui, F. Li, and K. Tomovic, "Hybrid symbolic-numeric framework for power system modeling and analysis," *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1373–1384, 2021.

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